

HOWARD CLARK and JOANNE HEAUME

2
BOOK

GENERAL SCIENCE TOPICS & ISSUES

FOR SENIOR STUDENTS



Quality ready-to-use
resources

GENERAL SCIENCE TOPICS & ISSUES

TOPICS FOR SENIOR STUDENTS

BOOK 2

Howard Clark and Joanne Heaume



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Introduction

General Science Topics & Issues presents sets of guided questions and activities on a number of diverse Science topics for Years 9 to 12 students. It is designed to inspire students to extend their understanding, vocabulary and ideas beyond the classroom lessons.

The topics in this two-book series cover each of the major sciences (Biology, Chemistry, Geology and Physics) and may help Year 10 students to

choose their senior Science for matriculation by highlighting their particular areas of interest and expertise.

The questions range from basic recall through comprehension and interpretation, analysis and research, so as to extend the learning capabilities of students. Activities such as crosswords and group discussion encourage initial interest in the more formal activities.

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Note: Topics 1 – 4 are found in Science Connections Book 1.

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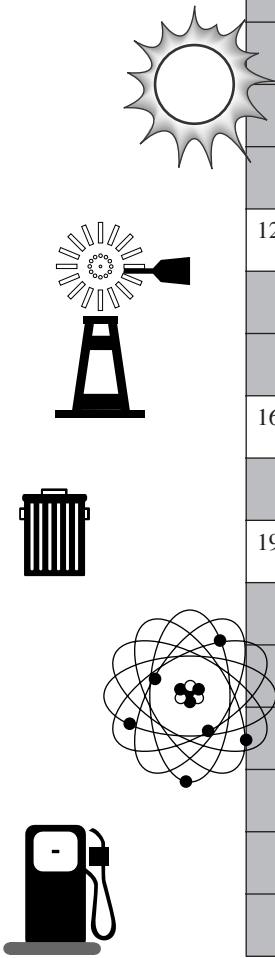
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Energy for Our Life

5a Energy Crossword



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WORDS

- alcohol
- atomic
- coal
- cold
- diesel
- energy
- fossil
- gas
- glucose
- heat
- hot
- hydro
- jog
- joule
- light
- nonrenewable
- oil
- petrol
- radiate
- renewable
- rot
- solar
- sound
- sun
- thermal
- tide
- waste
- KE, PE, mgh

CLUES

Across

- 5 Energy produced by moving water (short form) (5)
- 7 Energy we can hear (5)
- 8 This natural fuel is found near oil (3)
- 9 If something is hot it has ____ energy (4)
- 12 Stars ____ their energy through space (7)
- 14 Energy chemical produced by plants during photosynthesis (7)
- 15 Potential Energy (initials) (2)
- 16 If an object is not hot, it is ____ (4)
- 17 CO₂ released from burning is a _____ product (5)
- 19 Unit of energy (5)
- 22 We can see this type of energy (5)
- 23 This raw material is refined to produce petrol (3)
- 24 The energy produced by the sun (5)
- 25 Potential energy can be calculated by this formula PE = ____ (3)

- 28 Liquid fuel similar to petrol, used by trucks. (6)
- 29 Movement of the oceans due to gravity (4)

Down

- 1 This fast gait uses more energy than walking (3)
- 2 Coal is this type of fuel (6)
- 3 Fuel which is not replaced once used (12)
- 4 Capacity to do work, to cause some change (6)
- 6 Heat is produced when waste vegetation does this (3)
- 10 Energy produced from colliding atoms (6)
- 11 Fuel which can be replaced in a short time (9)
- 13 Liquid fuel produced by fermentation (7)
- 18 Liquid fuel for cars obtained from oil (6)
- 20 Another name for heat energy (7)
- 21 Solid fossil fuel (4)
- 24 Source of solar energy (3)
- 26 Opposite of cold (3)
- 27 Kinetic Energy (initials) (2)

Energy for Our Life

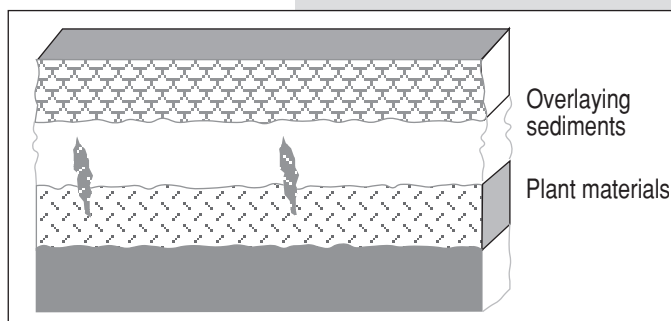
5b Sources of Energy

Our society uses energy from a number of different sources. You know that power stations burn coal, oil or natural gas. Some power stations use the movement of water to produce hydro-electricity and in some countries nuclear energy is the source for the production of electricity. Other sources of energy include solar, tidal and wind.

- 1 a List the steps to describe how a giant propeller on a hilltop allows us to obtain energy.
- b Name two advantages of this type of energy production.
- c List any disadvantages of this type of energy production.

Each of the different sources of energy can be labelled as either renewable, if the energy source can be replaced in the short term, or non-renewable where the energy source cannot be replaced in the short term.

RENEWABLE/NON-RENEWABLE FUELS



Non-renewable fuels include what we call 'fossil fuels'. Millions of years ago, areas of forest (in

the case of coal) or microscopic animals (to produce oil) were compressed under layers of earth as part of the sedimentation process in the cycle of rock formation.

The energy captured from the sun during the life of these organisms has been changed into the chemical energy of the compounds in their bodies. It is this energy that was stored until mined and then released in the form of heat and light. The amount of coal, oil and natural gas formed during these geological periods, although large, is finite or limited. Even if large quantities of coal were produced today it would be millions of years before this fuel could be utilised.

- 2 a What is a fossil fuel?
- b How is coal and oil produced?
- c Why do we call fossil fuels 'non-renewable'?

The reserve of useable fossil fuel is being reduced day by day and soon the total amount of energy available from this source will be zero. Scientists and politicians disagree over both the size of the reserves and the level at which they are being reduced. Fossil fuels also release carbon dioxide gas and ash when they burn. These wastes lead to global warming and other forms of pollution.

Other sources of energy, such as solar, wind or tidal energy, will be available for the lifetime of our Earth - that is millions of years - and are therefore considered to be continuous or renewable. The collection of wind, tidal and solar energy is also less harmful to the air we breathe because they do not release carbon dioxide or other waste gases.

- 3 a List and describe some of the disadvantages of using fossil fuels for energy.
- b What are some advantages of using coal and oil?
- c Is there any difference in the type of electricity produced by different fuels? Explain your answer.
- d Why is the use of wind, tidal or solar energy less harmful to the air we breathe?

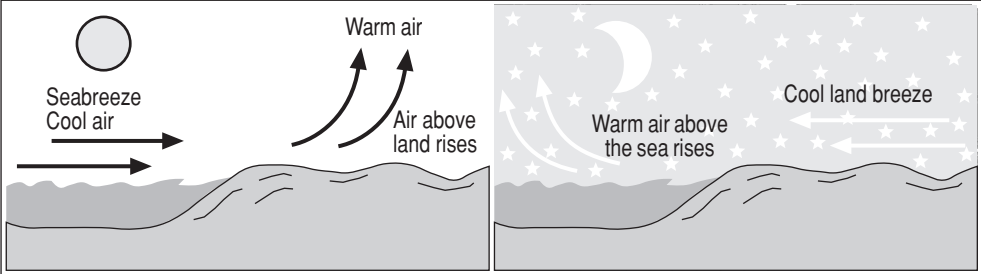
Generators and electric motors convert energy from one type to another to make it more useful to us.

- 4 a Is much energy wasted or lost in the conversion process? How can you tell?
- b Describe how and why energy conversions use up energy.
- c In small groups, develop flow diagrams to show the mechanism (process) for producing electricity from fossil fuel burning, nuclear, solar and wind/tidal energy production. Show where energy is lost in these processes.

Energy for Our Life

5c Energy from the Wind

Every second, energy from the sun hits the earth. It is this energy which can be changed into electricity either directly through solar cells or indirectly via the movements of air (wind) and water (tides and currents). Wind and tidal energy are examples of passive solar energy. Either way the energy is continuous and will remain until the end of the lifetime of the sun. Every day about 1.5% of the solar energy that hits the earth is changed into wind energy.



The use of wind farms causes reductions in carbon dioxide (CO₂), sulfur dioxide (SO₂) and nitrogen oxide (NO_x) compounds.

- 1

a

Name and describe three types of solar energy production.

b

Explain how wind energy and tidal energy are similar. Why are they called ‘passive’ solar energy?

c

What differences are there between wind and tidal energy production?
- 2

a

Describe what is happening in each of the two diagrams above.

b

Why does the breeze blow towards the land at the end of a hot day?

c

Explain what happens as the wind blows out to sea during the night.

With wind energy, the particles of air move fans connected to a generator. More information about how wind generators work can be found from the Australian Council for Renewable Energy website <http://www.phys.murdoch.edu.au/acre/refiles/wind/> or the US Department of Energy website <http://www.eren.doe.gov/windandhydro/>

The table below is from the British Wind Energy Association and it gives information about the wind farms in the United Kingdom in the month of October.

- 3

a

Calculate the average size of British wind farms in:

i)

turbines; and

ii)

MW of energy produced per farm.

b

Describe what you think is meant by ‘homes equivalent’.

c

Determine the average home equivalent for each turbine.

d

How many megawatts of power are consumed by the average home?

- 4

a

Is this reduction a good thing or a problem? Explain.

b

List some other changes that might be due to wind farms.

For more information about Australian wind power production research see the University of Newcastle website <http://www.wind.newcastle.edu.au/> or CSIRO <http://www.clw.csiro.au/products/windenergy/>

At the University of Newcastle Wind Group, research is mainly on small wind turbines. Some of the research carried out includes:

- Fatigue of turbine blades because of the higher rotational speeds of small wind turbine blades (compared to large turbines)
- The need to obtain energy even at low wind speeds
- Improving blade designs.

- 5

Imagine you are a researcher with the Wind Group at the University of Newcastle:

a

Choose one of the topics above and write in your own words what the research problem is.

b

Describe an experiment (or series of experiments) you could do to increase understanding in that particular area.

c

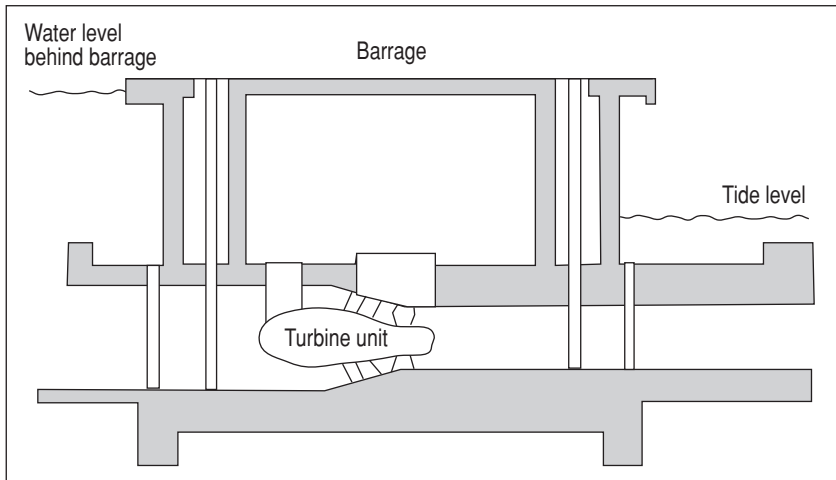
Present your idea as a proposal to send to the Wind Group.

Projects (wind farms)	Turbines	Megawatts (MW)	Homes equivalent	CO ₂ reductions	SO ₂ reductions	NO _x reductions
62	857	409	259,538	925,700 t	10,671 t	3126 t

Energy for Our Life

5d Energy from the Ocean

Energy from tides is produced the same way as wind energy but instead of a rotating fan, a turbine unit is used at the bottom of a barrage or dam. When the tide comes in water rushes past the turbine producing electricity. When the tide goes out the direction of the turbine is reversed and more electricity is made.

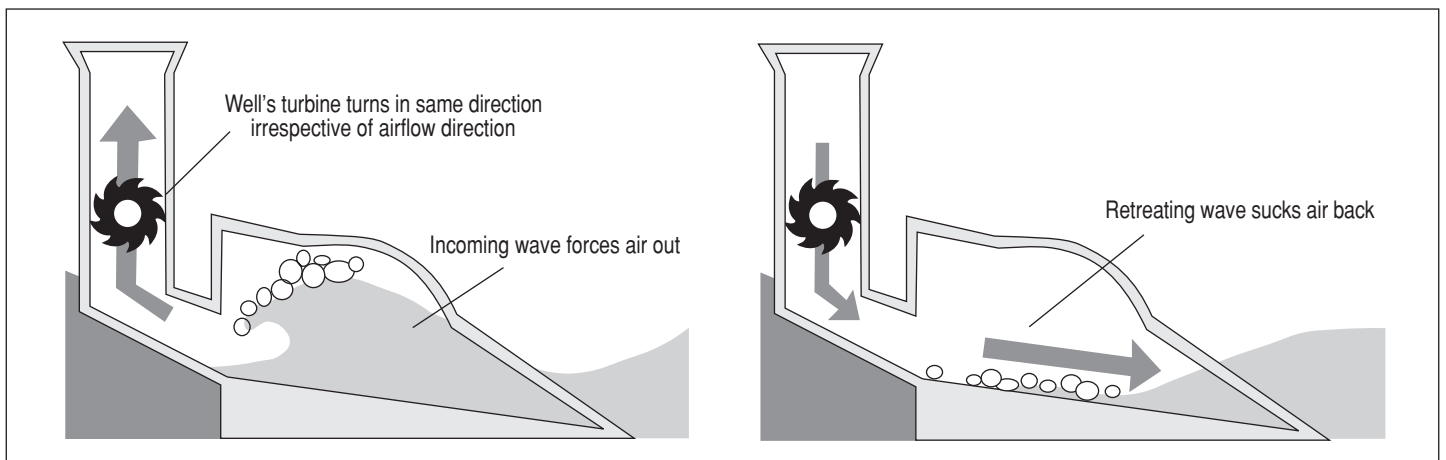


- 1**
 - a Describe what happens between low and high tide to produce energy.
 - b At what stage in the tidal cycle is there no tidal movement?
 - c When is there the greatest tidal movement?
- 2**
 - a What causes the tide changes around the world?
 - b What is the original source of energy that we collect as tidal energy?
 - c Is this source of energy renewable or non-renewable? Explain.

Even without large tidal movements, the waves crashing on a headland or shore can be used to collect energy. The two diagrams below show what happens when waves force air up a chimney.

- 3**
 - a What happens to the air in the chimney when the wave comes?
 - b How can this wave motion make electricity?
 - c What happens to the air when the wave moves back?
 - d How can the out flow of a wave also produce electricity?
- 4**
 - a List some of the advantages of wave power generation.
 - b Suggest some disadvantages of having these chimneys along the shoreline.
- 5**
 - a What causes waves in the ocean?
 - b What is the original source of energy that we collect as wave energy?
 - c Is this source of energy renewable or non-renewable? Explain.
- 6** Imagine you are designing a tidal power generator or a wave power generator.
 - a Describe some of the concerns or problems you might face.
 - b List some of the positive aspects of your design that would encourage your power source to be used.

Further information on tidal power can be obtained from Australian Renewable Energy website <http://renewable.greenhouse.gov.au/technologies/ocean/tidal.html>



Energy for Our Life

5e Electricity and Power

The energy from the sun is measured in units called joules (J). When this energy is used in an engine or to run some equipment it is called power and the unit of power is a watt. One watt (W) is one joule used every second (1 W = 1 J/s). For more energy and lots of power we use kilowatts (1000 W) which is 1000 joules used every second.

For larger amounts of power, particularly in household energy use, we use kilowatt hours (kWh). This unit measures the amount of energy used in an hour. As a result, 1 kWh is 1000 joules

for 3600 seconds (3 million 600 thousand watts). Sometimes energy is even measured in megajoules (1 million joules = 1 MJ).

In Summary

1 watt (W) = 1 joule (J) per second (s)
W = J/s
1 kW = 1000 watts
1 MW = 1 million watts
1 kW hour = 3 600 000 joules

Appliance	Power rating (watts)	Time used h/day	Energy use watts x hours	Cost/day 10 c/kWh	Cost per year
Refrigerator					
Freezer					
Microwave					
Toaster					
Oven					
Television					
Video					
Sound system					
Computer					
Printer					
Washer					
Iron					
Fan					
Air condition					
Heating					
Power tools					
Lights					
(other)					
(other)					
TOTAL					

- 1
- a

Write definitions for the following words: joule, watt, power, megajoule, megawatt.
- b

How many joules per second are used in an 800 W microwave oven?
- c

Many incandescent light bulbs are 60 W. How many joules of energy do they use per hour?
- d

A 1000 W room heater is left on for 8 hours. How many joules of energy are used?
- e

A 2500 W oven is used for 4 hours. How many kWh of energy are used?

Household power can be calculated using the table below.

- 2
- a

Complete the ‘Power rating’ column in the table for your home using the ratings listed on each appliance.
- b

Determine the average time used every day for each appliance and calculate the energy use per day.
- c

Calculate the cost of the appliance per day and per year. Use a general cost of 10 c/kWh.
- d

Add any extra appliances to the ‘other’ rows.
- e

Add up the totals for the energy and cost columns.

Use the information from your table to answer the next questions.

- 3
- a

Which appliance has the greatest power rating?
- b

Which appliance uses the greatest amount of power in your house?
- c

What percentage of your energy cost goes to this appliance?
- d

Mark the appliance in the table that you use most often. Comment on the energy use of this appliance.
- 4
- a

Is it possible for your family to reduce its energy costs by 10%? Explain how.
- 5
- a

The summer of 2004 had record high temperatures across Australia.
- b

Describe some ways you and your family keep cool in such conditions.
- c

Describe some alternative methods of keeping cool, which would use less energy.

Energy for Our Life

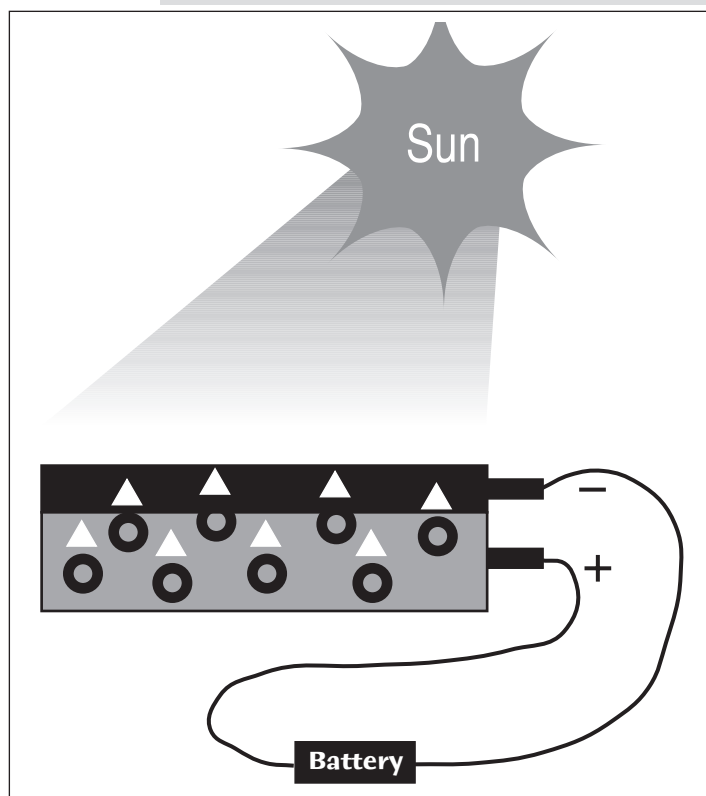
5f Solar Energy

We can use the power from solar energy in our own homes. Some solar panels collect the energy and transfer it directly into heat by warming water which runs through black pipes. These are commonly used in hot water services or to heat swimming pools. Solar energy can also be changed into electrical energy by solar cells.

- 1
 - a List any homes, businesses or activities where you have seen solar energy being used.
 - b Name and describe some advantages of using solar energy in the situations you noticed.
 - c Are there any disadvantages in using solar energy?
 - d How could solar-powered street lighting or public telephones in country areas be an advantage?

How do solar cells work?

Solar cells are usually made from thin pieces of silicon or other semiconductor material. When light hits the cell, some of it is absorbed within the semiconductor. This means that the energy of the absorbed light is transferred to the atoms within the semiconductor.



The energy knocks electrons loose, allowing them to flow freely. In the diagram the negatively charged electrons are shown as circles. The rest of the atoms are now positively charged.

- 2
 - a Describe how an electrical conductor is different from a nonconductor.
 - b List some examples of conductors.
 - c Name some nonconductors.
 - d How are semiconductors different from conductors or nonconductors?
 - e List some properties of a metal such as copper (Cu).
 - f List some properties of a non-metal such as sulfur (S).
 - g List some properties of the element silicon (Si).

Because of the way the silicon is organised in the cell, the electrons can only move in one direction (towards the upper dark colour in the diagram). The wire attached to this side becomes negative and the wire attached to other side, which has lost electrons, becomes the positive connection.

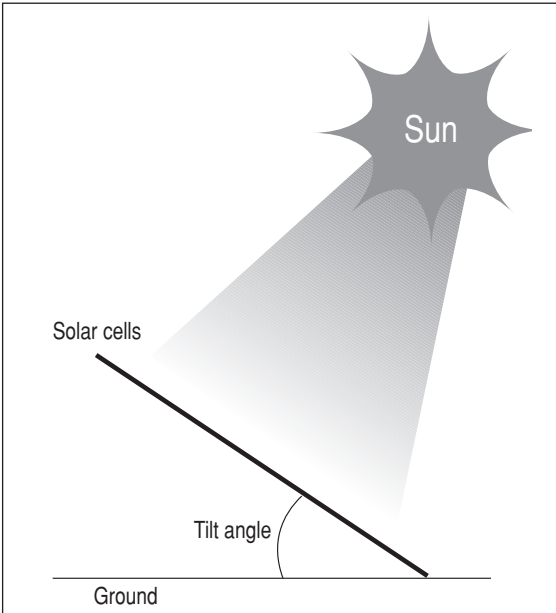
The electrons can only get back to the positive side of the cell through the wire. This movement makes an electrical current. Electrical currents can be used immediately, eg to run a radio or a calculator, or stored in a battery for use later on, as in street lighting. In the diagram a battery is shown where the energy can be collected.

- 3
 - a What makes electrons move in a solar cell?
 - b What element is mainly used for solar cells?
 - c Name and describe the three main particles that make up an atom.
 - d Explain why electrons travel through the connecting wires.
- 4 Draft a letter to your school Principal arguing that the school should invest in solar energy. Make sure you discuss the advantages and disadvantages to develop a valid and useful result.

Energy for Our Life

5g Using Solar Energy

The angle of tilt of solar panels greatly affects the amount of energy they receive at different times of the year. The diagram shows what is meant by the tilt angle. The following table shows the effect of different tilt angles on solar cells. The first row shows tilt angle from 20° to 70°. The different tilt angle means that different amounts of energy can be collected by the solar cell. The other rows show the maximum amount of solar energy falling on 1 square metre of solar panel in one day during the stated month. These figures are based on Sydney and the energy is measured in megajoules (MJ).



Energy falling on a solar cell at different angles of tilt						
Tilt angle	20°	30°	40°	50°	60°	70°
Sept	23.8	24.9	24.8	24.1	22.7	20.5
Oct	28.4	28.8	27.4	25.2	23.0	19.8
Nov	29.2	29.9	27.4	25.2	22.3	19.1
Dec	28.8	29.9	27.4	25.6	23.0	20.2
Jan	25.6	25.9	26.3	24.8	22.7	20.5
Feb	20.5	21.6	22.3	22.7	21.6	20.5

- 1
- a

What does the tilt angle do to the solar cells?
- b

Which angle would have the cells almost flat on the ground?
- c

Which angle and month collects the most energy? How many MJ per square metre are collected?

- d

Which angle and month collects the least energy? How many MJ per square metre are collected?

The tilt is necessary because the sun is never absolutely overhead. It is seen higher in the sky during summer months and lower in winter. In winter, a large tilt angle would be required because the sun is low and close to the horizon. The days are also shorter in winter, making it less useful for the collection of solar energy. In placing a panel in the best site, care must be taken so that trees or shade from neighbouring buildings do not decrease the efficiency of the solar cells.

Clear sky also means that a larger amount of solar energy can be collected. In the news and weather reports, the UV rating is often given. A high rating means that there has been a clear day with little haze or cloud to disturb the solar radiation. Some clouds, however, do allow radiation through and as a result, the solar cells gain energy even if it appears to be an overcast day. People can also very easily get sunburned.

- 2
- a

Using the grid or graph paper, draw a line graph for the energy collected each month for 30°.
- b

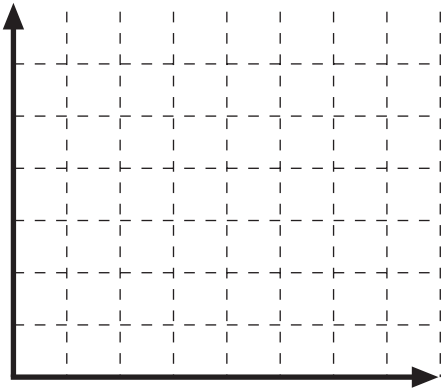
Repeat the graph using different colours for each angle.
- c

Which angle gives the best result?
- d

What is the latitude (° south of the equator) of Sydney?
- e

Comment on any relationship between the best tilt angle and the latitude of Sydney.
- f

Find out the latitude of your town or capital city. What would be the best tilt angle for solar cells at this place? Explain why.



Energy for Our Life

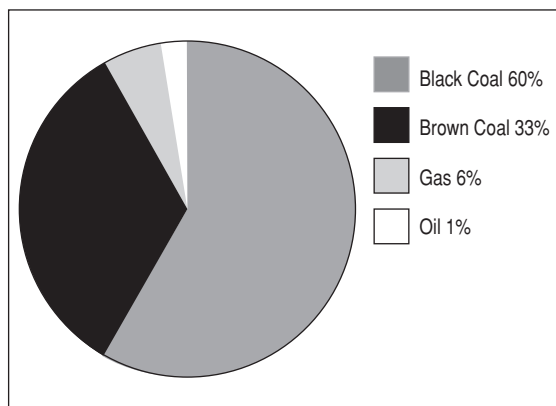
5h Chemical Energy

Most of our energy is produced by the burning of coal and oil. These substances store as chemical energy the sun's energy from millions of years ago. Examples of fuels refined from coal and oil are natural gas, kerosene, petrol and ethanol.

The process of burning, more correctly called combustion, is a chemical reaction between a fuel and oxygen producing energy in the form of heat and light. Combustion releases the waste products carbon dioxide and water as well as energy. If complete combustion takes place, this is all that is produced. Sometimes incomplete combustion occurs and the waste products include ash and other particles which have not been fully burnt. This means that some fuel is wasted.

- 1
 - a Where does the energy in coal and oil originate?
 - b Why are fossil fuels called non-renewable?
 - c List some examples of fossil fuels and describe where they are used.
- 2
 - a List the two raw materials needed for combustion.
 - b What is produced during complete combustion?
 - c Describe the difference between complete and incomplete combustion.
 - d What extra wastes are produced during incomplete combustion?

The graph below shows the types of fossils fuels used for electricity generation in Australia.



- 3
 - a Which fossil fuel was used for most electricity production in Australia?
 - b Australia imports oil. What is it used for if we only generate 1% of our electricity from oil?
 - c Redraw the graph in bar graph style.
 - d What is the difference between black and brown coal?

There are some major problems with the use of fossil fuels as sources of energy:

- They are being used up and one day will run out completely. The current rate of use is far greater than the production rate and the needs for energy are increasing every year.
- Combustion produces waste products, including dust and ash pollution which can cause lung diseases as well as wasting energy.
- Global warming caused by carbon dioxide (CO_2), the main waste product. CO_2 in the atmosphere allows the sun's light through to heat up the surface of the Earth during the day but does not allow the heat to escape from the Earth at night. The average temperature of the Earth is increasing and it is thought that low lying and coastal areas could be flooded from the melting of the polar ice caps if the temperature becomes too high.
- Wasted energy. The heat from burning fossil fuels is not all converted into usable energy. There is much wasted energy and unfortunately this means that more fuel has to be used to supply our requirements.

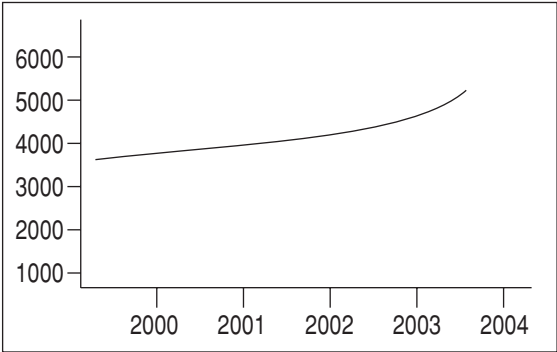
- 4
 - a What is the problem with coal and oil being non-renewable energy sources?
 - b Describe how dust and ash might cause lung cancer.
 - c Name the most common cause of lung cancer.
 - d What is the cause of global warming?
 - e Suggest some possible results of global warming.
 - f Suggest ways we can reduce the problem of global warming and reduce the amount of wasted energy.

Energy for Our Life

5i Energy Consumption

According to the Australian Bureau of Statistics (ABS), the total amount of energy used in Australia each year has risen by over 60% since 1980. This increase in energy is due to the growth of both our population and economic development.

We have become more efficient, however, as the amount of energy used per person has increased by only 24% over the same period of time. The graph shows the energy use in Australia in petajoules over the last few years (1 PJ = 1 000 000 MJ).



- 1
- a

How much more energy do Australians use today compared to 20 years ago?
- b

How much more energy does each Australian use compared to 20 years ago?
- c

Describe in your own words the terms *petajoule* and *megajoule*.
- d

How much energy was used in Australia in 2000?
- e

How much energy do you anticipate being used in 2004?

- 2
- a

Compare the change in energy consumption in the last 10 years in Australia and Vietnam.
- b

Suggest reasons for the large difference.
- c

Which country has the largest energy consumption per person?
- d

List some reasons why this is so.
- e

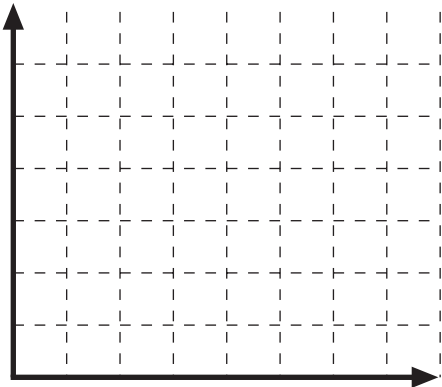
Which country has the largest total energy consumption?
- f

If China (population 1200 million) or Vietnam (pop. 77 million) used as much energy per person as Australia or the USA, how much would each country use in total?

- 3
- a

Using the blank graph or your own paper show the per capita (per person) consumption of energy for each country.
- b

Present a separate graph showing the change in the last 10 years for each country.



The table below shows energy consumption in Australia and a number of other countries.

Energy Consumption	Australia	Canada	USA	France	UK	China	Japan	Indonesia	Vietnam
Millions of MJ	2767.9	7616.6	60,434.8	6783.5	6798.9	25,685.6	14,112.9	2339.6	522.9
(%) Change in last 10 years	31.1	19.9	12.6	19.1	15.1	45.5	34.6	117.9	221.1
Consumption per person (1000 MJ)	151.3	254.2	227.6	116.2	115.7	21.1	112.1	11.9	6.9

Energy for Our Life

5j Research Project

In this topic a number of websites have been listed. There are many other sources of information on energy. In particular, the websites listed below are very useful.

Australian Council for Renewable Energy
<http://www.phys.murdoch.edu.au/acre/refiles/wind/>

Australian Home Energy Guide
<http://www.greenhouse.gov.au/yourhome/home.htm>

Australian Renewable Energy
<http://renewable.greenhouse.gov.au/>

Australian Wind Energy Association
<http://www.auswea.com.au/>

British Wind Energy Association
<http://www.britishwindenergy.co.uk/>

Californian Energy Story site
<http://www.energyquest.ca.gov/story>

CSIRO
<http://www.clw.csiro.au/products/windenergy/>

Envirolink
<http://www.envirolink.org/>

International Council for Local Environmental Initiatives
<http://www.iclei.org/efacts/>

Murdoch University Remote Area Power Supply
<http://www.phys.murdoch.edu.au/WebRAPS/>

Ontario Power Generation (Canada)
<http://www.opg.com/default3.asp>

Regional Biomass Energy Program (USA)
<http://www.ott.doe.gov/rbep/>

Renewable and Sustainable Energy
<http://solstice.crest.org/>

Sustainable Energy Authority of Victoria
http://www.seav.vic.gov.au/renewable_energy/index.html

US Department of Energy
<http://www.energy.gov/engine/content.do>

Imagine you are a designer employed by both the State Government and the local mining company. Your job is to develop housing, schools, shops and other services for the workers in a new mine in a remote part of the state.

TASKS

Task 1 – Describe the different factors of the environment that need to be considered before producing a design.

Task 2 – Construct a table to list the advantages and disadvantages of the different types of energy that could be used by the mining company (in the mining process) and the residents of the proposed town.

Task 3 – Using the information from Task 1 and Task 2, design housing services that take account of the extremes of climate and yet provide comfortable, safe and affordable housing for the workers and their families.

Requirements

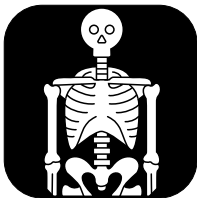
- 1** The mining company wants the most economical solution yet with a quality that will encourage workers to stay with their families for the duration of the long-term mining project.
- 2** The Union requires safe and comfortable housing for the miners with low-cost energy bills even though the local climate is extreme and energy use may be high.
- 3** The State Government and Local Council require all appropriate safety and standards conditions to be met.

Situation

The mine is situated in a tropical area with regular long periods of sunlight and steady wind. Closest access to the electricity grid is many hundreds of kilometres away and road transport of oil or coal is very expensive. There is a river in the region but it only flows during the wet season.

Debate If different groups in your class present alternative conclusions, present them in a debate to obtain levels of support within your class community.

6a Bionics Crossword



1		2													
										3					
	4					5				6			7		
8															
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					10										
				11											
							12								
		13													
		14								15					

WORDS

- alveoli
- aorta
- bicuspid
- diastolic
- heart
- hinge
- keyhole
- sinoatrial
- superalloy
- thermograph
- trachea
- xray
- UHMWPE
- ultrasound
- valve

CLUES

Across

- 1 A non-invasive medical technique that uses sound waves to form an image. (10)
- 3 The main artery which carries the blood from the heart. (5)
- 4 A non-invasive medical technique that uses radiation and forms a black and white film, often used to determine if a bone is broken. (4)
- 5 Type of valve found in the heart. (8)
- 9 The substance to assist with the lubrication of an artificial hip joint; the acronym for ultra high molecular weight polyethylene. (6)
- 11 Organ of the body that pumps blood. (5)
- 13 Found in the lungs, the place where the oxygen is diffused into the blood. (7)
- 14 Otherwise known as the windpipe. (7)
- 15 Type of joint and its movement is similar to a door hinge. (5)

Down

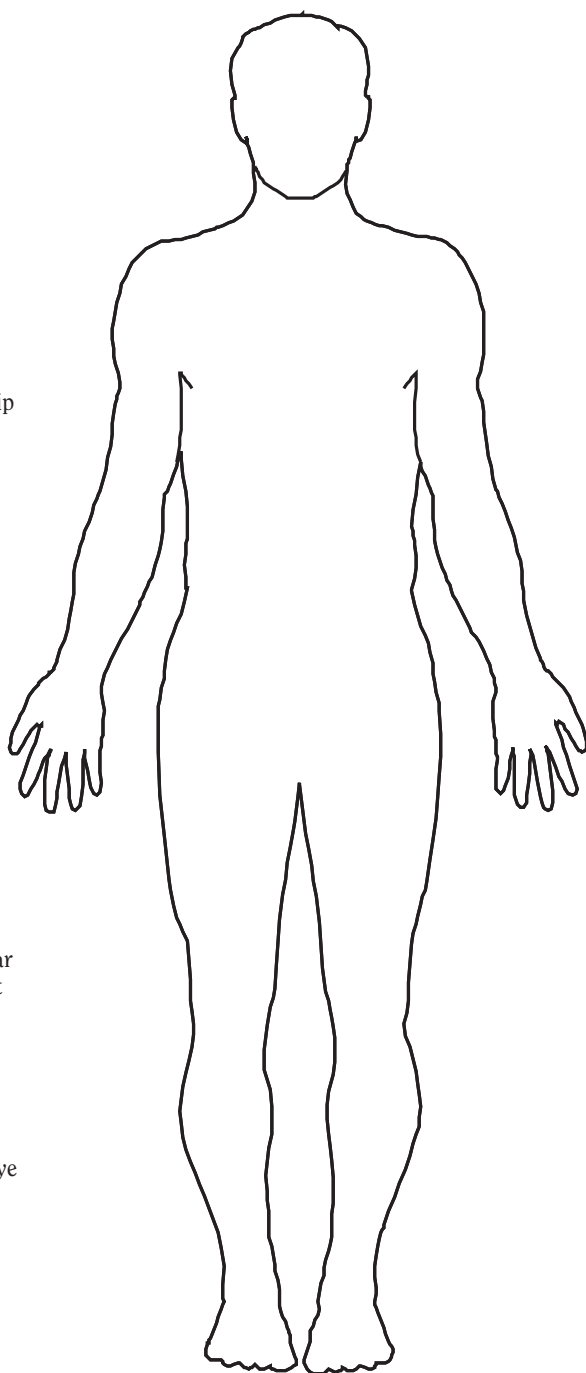
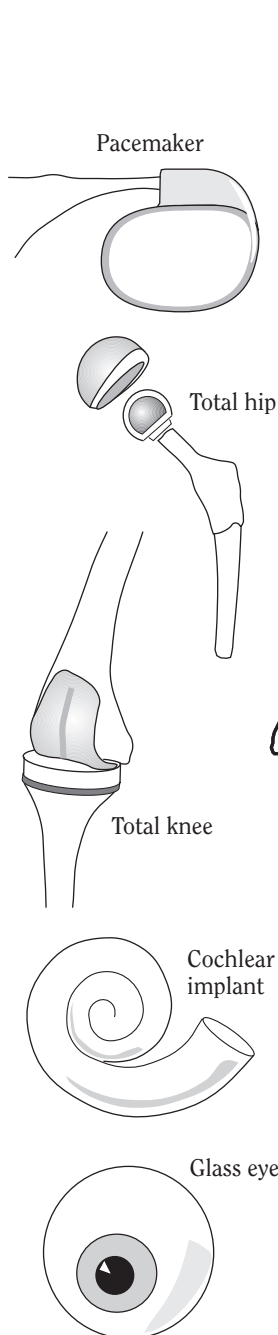
- 2 Produced from a non-invasive medical technique that uses infrared imaging cameras. (11)
- 6 Combined metal, good for replacing ball and stem in a ball and socket joint. (10)
- 7 Pressure measured when the ventricle of the heart has relaxed and fills with blood. (9)
- 8 Where the natural pacemaker is located in the human body. (10)
- 10 Form of minimal invasive surgery that uses small incisions only. (7)
- 12 Found in the heart and separates the atrium from the ventricle. (5)

6b Biomaterials

BIOMEDICAL DEVICES

Biomedical devices or implants are artificial devices that perform the same function as a certain part of the body. For example contact lenses for focusing light on the back of the retina; or pacemakers that give electrical impulses to the heart to ensure continual pumping of blood.

- 1** The diagrams to the left are biomedical devices. Draw a line to show where each of these devices would be located in the body. Write a label to state its function.



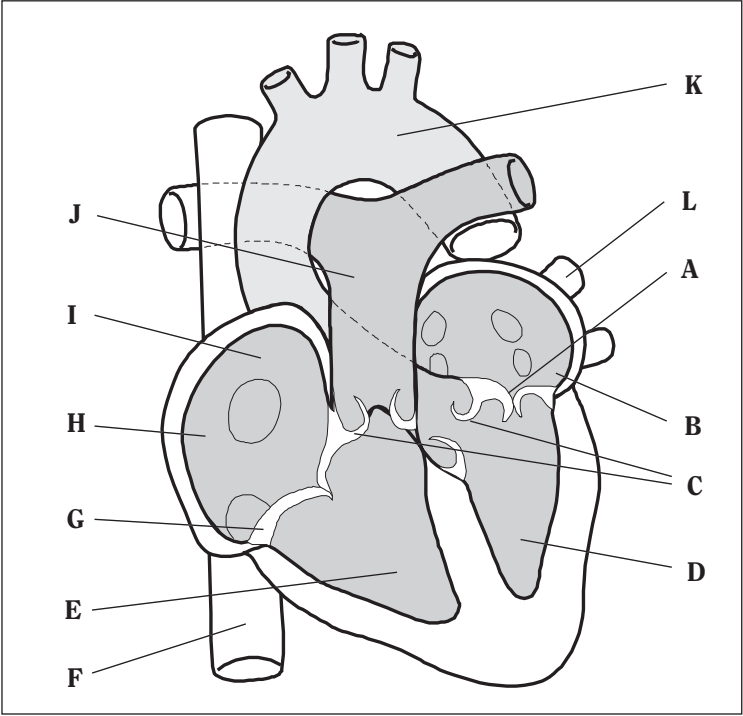
BIOMATERIALS

Biomaterials are materials or substances that are used to make biomedical devices in the body. For a biomaterial to be chosen it must have certain characteristics that suit the area or body part it is going to replace. All biomaterials must be biocompatible so not to interfere with the body chemistry.

- 2** The following list contains biomaterials that are used in the body. For each biomaterial, state where it is used and why it is effective in that area.
 - a** superalloy
 - b** ultrahigh molecular weight polyethylene (UHMWPE)
 - c** silicone
- 3** Using the Internet or library, research each of the following key words in relation to biomedical devices and answer the questions below:
 - Pins, screws and plates
 - Artificial joints
 - Artificial valves
 - Pacemaker
 - Dentures
 - Lenses
 - Prosthetic limbs
 - Cochlear implants
 - a** Write a short description for each device. Ensure you:
 - state what biomaterial each device is made from; and
 - explain the exact function of each biomedical device.
 - b** Collate all this information into a timeline showing when each biomedical device was developed and first used.

The heart is one of the vital organs in the human body. Its sole function is to pump blood around the body. One side (the right-hand side) of the heart pumps blood to the lungs where the blood becomes oxygenated. The left side of the heart pumps blood to the rest of the body.

- 1
- Describe why the left side of the heart is thicker than the right side of the heart.
- 2
- What is the function of the coronary artery?
- 3
- The list to the right contains names of the parts of the heart. Use these labels to label the diagram of the heart.
- 4
- For the parts of the heart that you have labelled above, complete the table below that shows the structure and the relative function.



- 5
- a

Explain what the symptoms would be for someone who had a hole in the central wall of the heart, between the left and right ventricle.

b

Describe the reason for each of the symptoms you list.

Part	Structure	Function
Aorta		
Left ventricle		
Right ventricle		
Left atrium		
Right atrium		
Sinoatrial node		
Vena cava		
Pulmonary artery		
Pulmonary vein		
Bicuspid valve		
Semilunar valves		
Tricuspid valve		

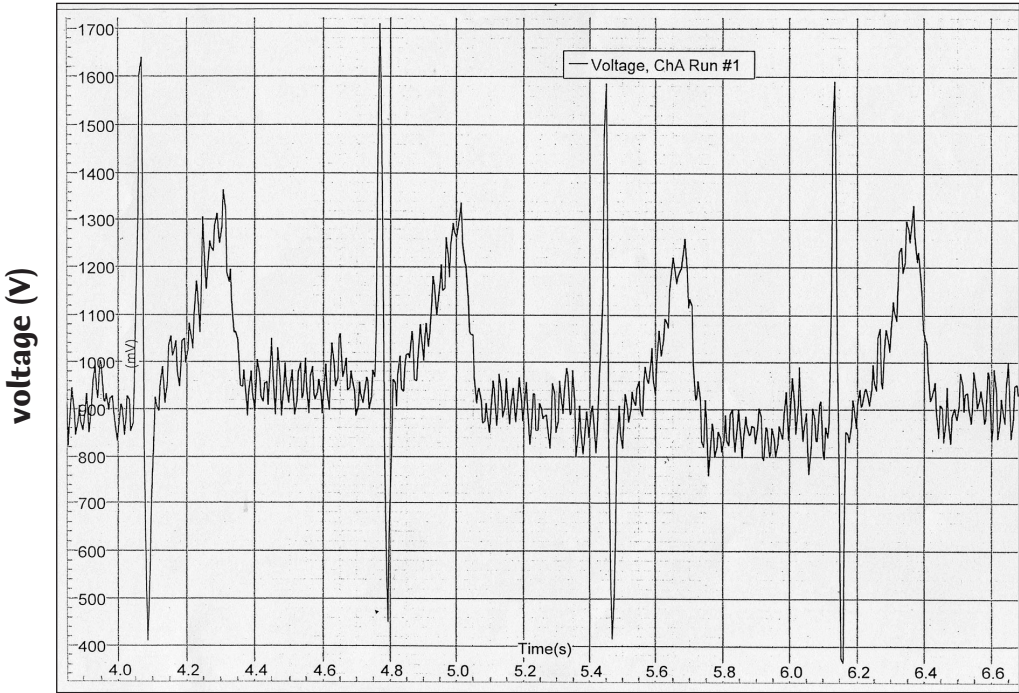
6d ECG - Electrocardiographs

The heart pumps because electrical impulses are sent from the sinoatrial node – the natural pacemaker – which makes the heart pump. An electrocardiograph records the electrical activity of the heart. Sometimes disease, age or lifelong activities can affect the structure and the muscular response of the heart.

- 1 When might it be essential to have an electrocardiograph?
- 2 Opposite is an example of an electrocardiograph. When doctors analyse electrocardiographs they look for specific parts. On the electrocardiograph below draw arrows to the parts the doctor would be looking for.

Doctors would look at an ECG similar to that above.

- 3 For the above diagram complete the two analysis tables.



a

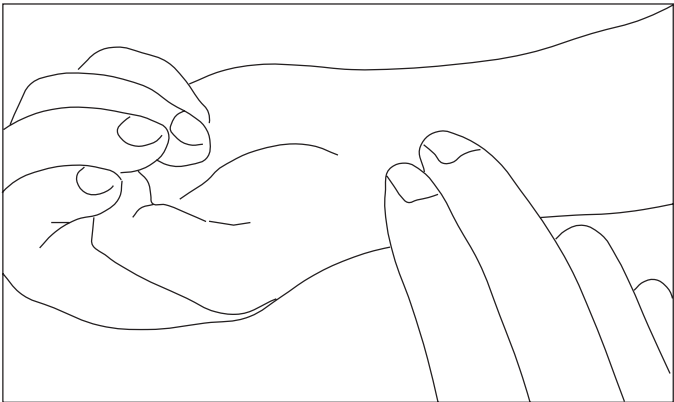
Item	Time (s)
P wave begins	
Q wave begins	
R wave begins	
S wave begins	
T wave begins	

b

Item	Time	Typical Time
P-R interval		0.120 - 0.200 s
QRS interval		< 0.1 s
Q-T interval		0.380 s

The ECG can be used to look at the electrical impulses of the heart and determine if there are any abnormalities of the heart. The heart can also be monitored by determining your pulse. This can be done by running your finger down the line of your thumb to your wrist and feeling for the pulse; count the number of beats for a 15 second period and multiply this value by four. This will give you the beats per minute (bpm).

- 4 When a patient's heart begins to fibrillate they will often be treated with an electrical shock. Why are patients treated by this method?



6e Pacemakers and Heart Valves

HEART VALVES

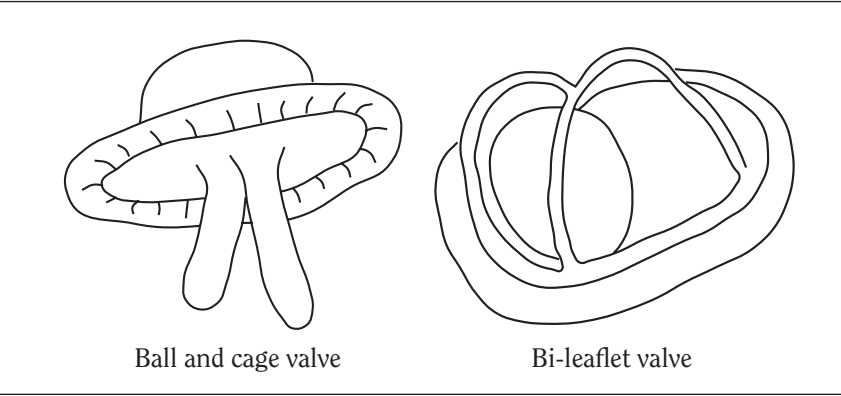
The heart contains many valves that allow the blood to flow in only one direction and to prevent backflow. The bicuspid valve is found in the left side of the heart between the atrium and the ventricle and it has two points or attachment areas. The tricuspid has three attachment points and is found in the right side of the heart. The semilunar valve is different again as it covers the opening of the pulmonary artery.

- 1 Distinguish between the following valves that are found in a working human heart. (Hint: the names are used to describe the structure of valves.)
- a Bicuspid valve

b Semilunar valves

c Tricuspid valve

Heart replacement and heart part replacement have been areas of research for many years. One example is artificial valves. The diagrams below show two types of artificial valves that have been used to replace heart valves.



- 2 Describe how the two artificial valves illustrated above work.
- 3 Identify what these artificial valves could be made of.
- 4 Research activity: Research the history of involved with the invention and development of heart valves. Represent this information in the form of a timeline, describing each of the findings.

PACEMAKERS

At times the heart does not beat correctly. It is either too fast or too slow, and this is when the heart needs an artificial pacemaker. Pacemakers are small electronic devices that replace the sinoatrial node of the heart. They produce small electrical impulses that allow the heart to beat regularly. The general term of arrhythmia, or abnormal heart beat, can be classified into *bradycardia* (slowing of the heart beat) and *tachycardia* (speeding up and quickness of the heart beat).

The speed of the heart can determine the pressure in the heart. This is also affected by the width of the arteries (narrow or wide) through which the blood is pumped. When the ventricle relaxes the blood pressure reading is called *diastolic*, and the *systolic* pressure reading is when ventricles contract and blood is pumped into the arteries. Any change in the heart rate affects the flow of oxygen to the body and we could lose breath, become tired or show symptoms of lack of oxygen by parts of our body, in particular the lips and ends of the fingers, turning blue.

- 5 Explain the following terms relating to an interruption in the normal rhythm of the heart.
- a Arrhythmia

b Bradycardia

c Tachycardia

d Diastolic

e Systolic
- 6 What are the symptoms of someone who is suffering from arrhythmia?
- 7 When someone is suffering from some type of severe heart rhythm problems a paramedic may resort to using paddles to give the heart an electric shock. Describe why the paramedic does this and what effect it has on the heart.

6f Skeleton and Joints

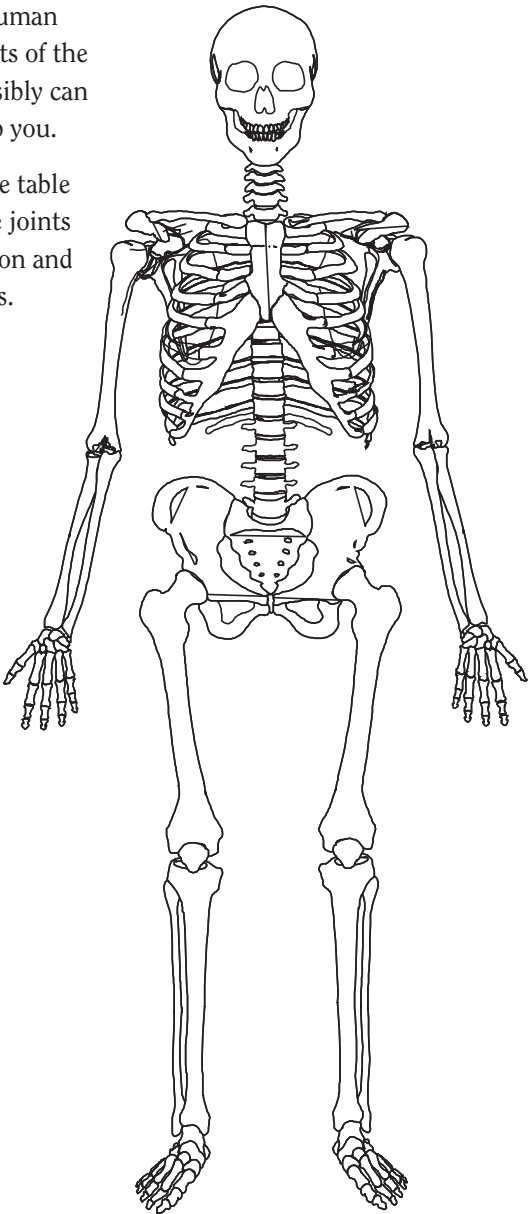
- 1

Below is a diagram of the human skeleton. Label as many parts of the human skeleton as you possibly can using the word bank to help you.
- 2

Using the information in the table below, identify and label the joints in the diagram of the skeleton and state which type of joint it is.

Word bank

Pelvis
Clavicle
Skull
Humerus
Femur
Lumbar vertebrae
Patella
Tibia
Tarsals
Carpals
Ulna
Metatarsals
Metacarpals
Sternum
Radius
Rib
Fibula



The joints in the human body allow for the body to be able to perform many tasks and to be able to use tools.

Joint type	Movement	Example
Ball and socket	Movement in many directions. Able to perform circular rotation.	Leg into pelvis joint at hip
Hinge	One plane only. Backwards and forwards similar to a door hinge.	Elbow
Double hinge	Backwards and forwards as well as side to side.	Base of thumb
Sliding	Allows gliding movements in all directions.	Joints in spinal bones
Pivot	Side to side only.	Where head is attached to spine

- 3

The most common biomaterials used in joint replacement are: super alloys, ultrahigh molecular weight polyethylene (UHMWPE) and silicone. For each joint decide the biomaterial that could best replace the joint.

6g Minimal Invasive and Surgical Techniques

Surgical techniques are often required to treat patients of their ailments. Today both full surgical and minimally invasive surgical techniques are used.

- 1 Discuss the advantages and disadvantages of using full surgical and minimally invasive surgical techniques for treating patients.
- 2 Present the data from the table in a graph.

Operation	Stay in hospital (days)	
	1970s	Current day
Appendicitis	5	1 - 2
Bypass surgery	14 - 21	approximately 5
Tonsillitis	1 - 2	1 - 2
Hysterectomy	21	approximately 5

- a What trends can you see from your graph and from the information in the table?
- b Explain any differences you may notice in hospital procedure in the 1970s compared with today.

An example of a minimally invasive technique is keyhole surgery. This type of surgery involves two or three very small incisions and many specialised instruments including a laparoscope, which inspects and provides clear vision of the abdominal area. The gastroscope is a specialist endoscope which inspects the gullet and stomach. A fiberscope is a specialist device that contains optical fibres for viewing the internal organs. Lasers are also used to help to reduce bleeding.

- 3 Define the following terms:
 - a laparoscope
 - b gastroscope
 - c endoscope
 - d endoscopy
 - e fiberscope
- 4
 - a List the types of medical procedures that now use keyhole surgery.
 - b Describe how keyhole surgical techniques would be conducted.

- 5 Research keyhole surgery using keywords such as:
 - Endoscope
 - Laparoscope.
- a Draw a diagram of instruments that could be used to conduct keyhole surgery and for each instrument describe how it works.
- b Design your own surgical instruments. Describe the problems they could solve and suggest ways in which this might advantage the relevant surgical procedures.

Approximately 20 years ago, keyhole surgery and the instruments involved did not exist. Instead, patients needed to be treated using open surgery where the doctor would make a full incision to be able to manipulate the organs. There were, and still are, many problems with this type of surgery, such as infection and time of recovery due to the initial incision.

- 6 List some of the problems of open surgery and describe how these are reduced using keyhole surgery.
- 7 Use the internet to find figures and data to support or refute this statement about open surgery.

‘The discovery of keyhole surgery and optical fibres has allowed humans to have a greater understanding of the way our body works.’

- 8 Considering this statement, discuss how advances in technology have had an impact on our understanding of how the body works.

6h Non-invasive Medical Techniques

Non-invasive medical techniques involve any form of medical technique that does not penetrate the skin. Such techniques are generally used as diagnostic tools, eg to determine if a tumour is present or if a bone is broken. Judgement on how to treat the injury will then be made based on the information. Often different techniques are used to give an overall ‘picture’ of the problem.

- 1 Identify the advantages and disadvantages of using non-invasive medical techniques.

Types of non-invasive medical techniques include: x-ray, often used to determine broken bones; PET, which can provide images of fluid flow (such as blood) through the body; thermography, using infrared images which can show increased blood flow to a certain area; and MRI, which can show detailed images of body tissue taken from any angle and through any section.

Non-invasive techniques require no entry into the body, whereas minimally invasive techniques do use some, even though small, entry into the body.

- 2 Explain why and how each of the following techniques are used.
- a X-rays
 - b Ultrasound
 - c Positron Emission Tomography (PET)
 - d Thermography
 - e Magnetic Resonance Imaging (MRI)

MAGNETIC RESONANCE IMAGING (MRI)

MRI has a number of advantages and, like every procedure, a number of disadvantages. These are listed in the table below.

Advantages of MRI	Disadvantages of MRI
Does not use ionising radiation.	Unable to treat people with a pacemaker or inserted electronic device as the magnetic field would interfere with its workings.
Low incidence of side effects.	Patients often feel claustrophobic.
No biological danger to humans with magnetism or magnetic fields associated with MRI.	Machines are very noisy.
Can form images in any plane of the body and at any angle.	Patients need to stay still for long periods of time.
	MRI equipment is very expensive.

- 3 Draw up similar tables listing the advantages and the disadvantages for the following non-invasive medical techniques: x-rays, ultrasound, positron emission tomography (PET), thermography.
- 4 How do non-invasive medical techniques differ from minimally invasive medical techniques? Contrast the two techniques and give an example of each.
- 5 Describe a medical problem you may know about from personal or family experience or that you may have heard about on television. Describe any procedure that was used. Could the patient have been helped by using non-invasive techniques? Explain.
- 6 Using the table below, match the following terms with the correct technique.
- Magnetic Resonance Imaging (MRI)
 - X-ray
 - Computerised tomography (CT) scan
 - Thermography
 - Ultrasound

Area of the body requiring analysis	Non-invasive diagnostic technique
a Leg	
b Head	
c Uterus	
d Liver	
e Breast	

6i The Respiratory System and Respirators

Respiration is the process of using oxygen to release the energy of our food. Commonly, however, the term refers to the obtaining of oxygen by the body so that it gets to where the food is in the cells. To do this properly, the body needs an efficient means of getting air into the body structure by the nasal cavity or nose, through the trachea and bronchi into the microscopic structures of the lungs. The pulling in of fresh air and the releasing of used air is done by specific muscles around the ribs and in the diaphragm.

- 1 Label the following parts of the respiratory system on the diagram:
- a Nasal cavity

b Alveoli

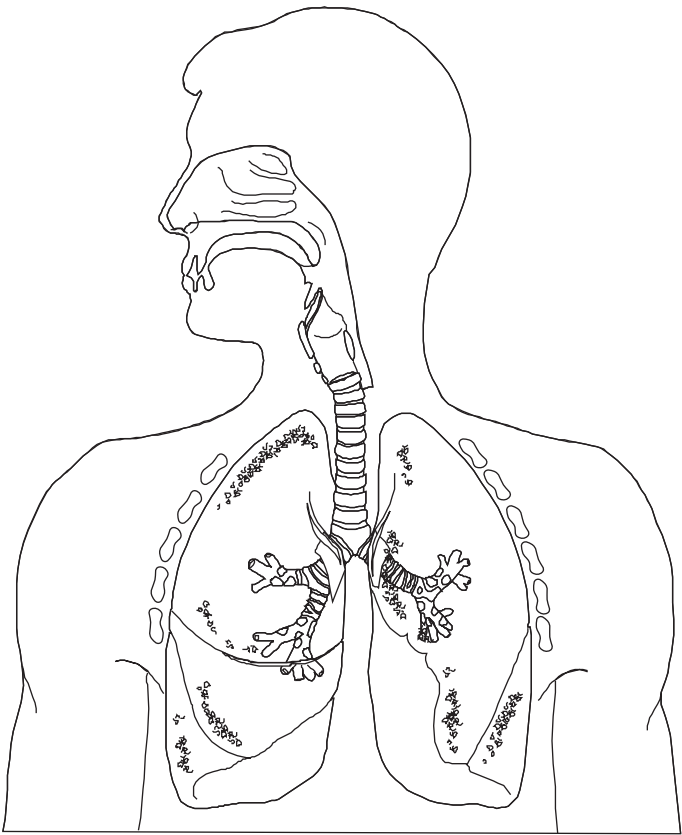
c Lungs

d Intercostal muscles

e Bronchi

f Trachea

g Capillary network around alveoli
- 2 Construct a table that shows the function for each of the parts that you have labelled.



Structure	Function

There are certain diseases that prevent the lungs from operating correctly. Polio is an example of such a disease. It causes paralysis of muscles including the intercostal muscles, vital for breathing. In the 1950s, children with polio were placed in an ‘iron lung’, a **negative pressure ventilator**. This machine worked by sealing around the body of the child then the pressure inside decreased and increased to allow for normal breathing to occur. When the polio epidemic was at its highest it was not uncommon to see whole wards of patients in iron lungs.

Nowadays, with an increase in knowledge, iron lungs are no longer used. Instead **positive pressure respirators** or ventilators are used. These are a lot more compact and work differently to the negative pressure ventilator of the iron lung.

- 3 Imagine that you are a medical sales person and you are to visit local hospitals and promote the latest positive pressure respirators or ventilators. Your job is to construct a pamphlet that can be used to promote these respirators or ventilators. You may decide to compare them favourably to the iron lung.
- 4 Certain types of operations require the use of an artificial lung. State which type of operations would require an artificial lung and for each instance explain why?

6j Research Project

CLONING BANKS

Imagine that a friend of yours was caught in a house fire. She is okay, but during the fire her ear was badly burnt. It is having trouble repairing and it will never grow back. The ability to grow another ear using cloning banks is a viable option.

Research what is meant by a 'cloning bank'.

- 1 Describe how cloning banks could be used in this example above.

Now think of the ETHICS.

- 2 How far could we go with cloning banks?
- 3 How far should we be allowed to go with cloning banks?
- 4 How could we control how far we go with cloning banks?

Decide where you stand in regard to cloning banks and whether you think they should exist and how they should function.

OPERATIONS FROM ACROSS THE WORLD!

- 5 Make a flow diagram to represent the transfer of information that would take place in the following situation:

Priya has a rare brain disorder and the surgeons in her part of India are not experienced in dealing with this situation. They have scheduled an operation to take place tomorrow and there is a specialist on call in Australia to assist.

The two doctors need to be able to communicate to allow for the operation to be a success.

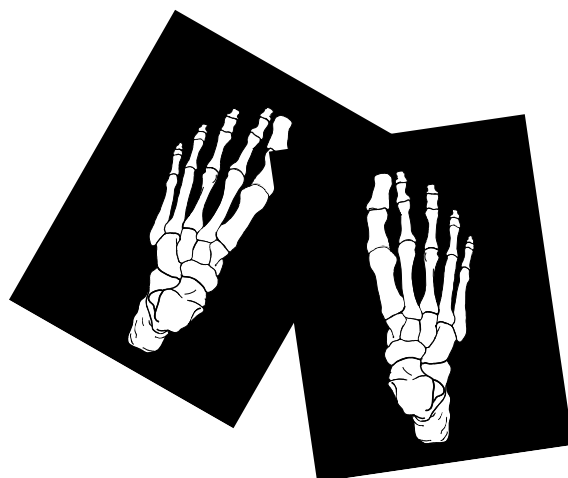
In your diagram include all information systems and devices used. Explain how each information system works.

- 6 Describe how this procedure would have taken place 15 years ago.
- 7 Make assessments on how scientific advances have contributed to the development in this type of situation.

- 8 Make an assessment on how this information impacts on society and the environment.

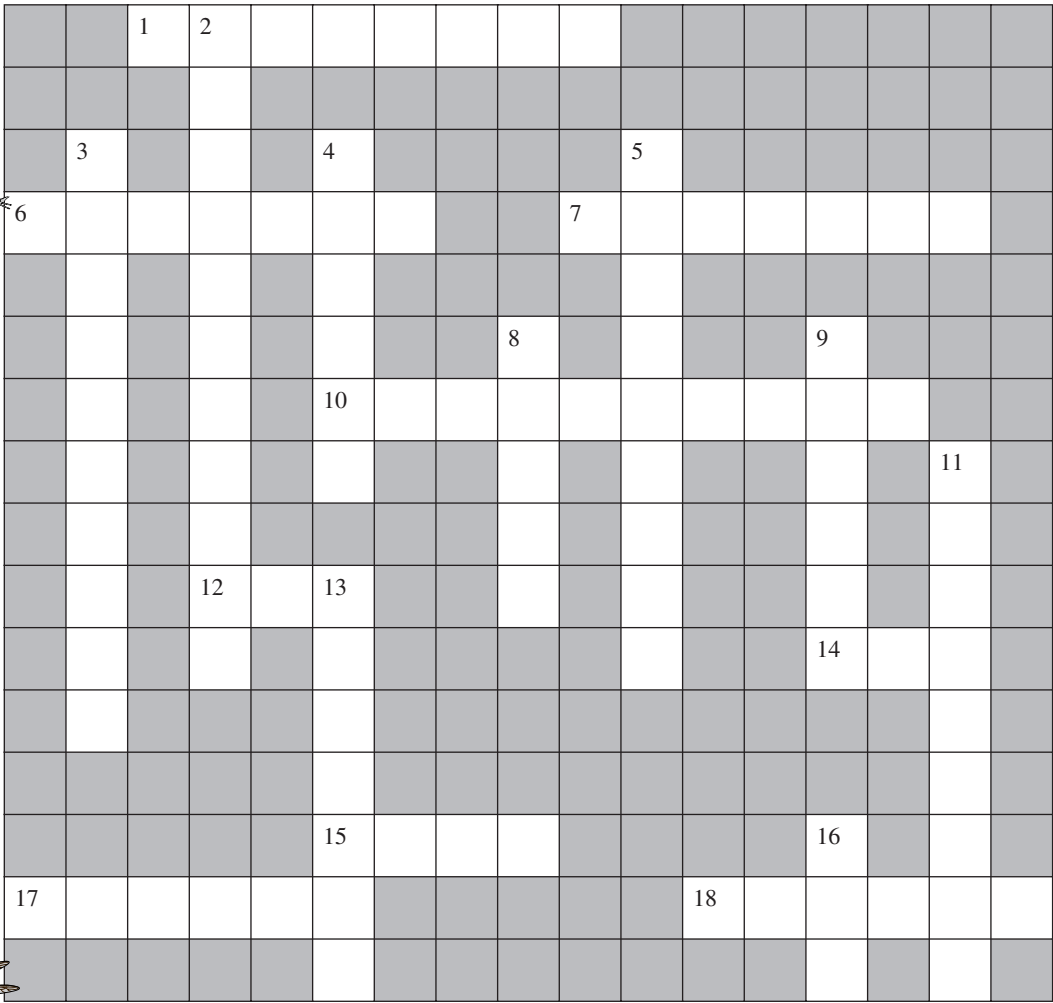
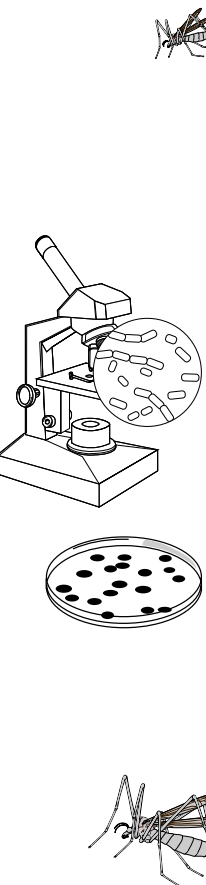
METHODS AND TECHNOLOGY

- 9 Investigate how death in surgery has decreased due to minimal invasive techniques and using non-invasive techniques as a diagnostic tool. Present data to support your conclusion.
- 10 Suggest what effect this might have on society. Include in your investigation how the stay in hospital has decreased allowing for more beds and whether this has any effect on patient care or recovery rates or financial costs to society.
- 11 The use of X-rays has been controversial. At the time of its discovery they were very popular and were even used to determine the size of peoples' feet when they were buying shoes. The person being fitted for shoes would stand on a platform and an X-ray of their feet would be taken. We are now more aware of the side-effects of being exposed to X-rays and use them only for certain diagnosing medical conditions, and even then strict safety procedures are employed to limit exposure.
 - a Research X-rays and construct a timeline of their development and use.
 - b Investigate the maximum level of radiation that a patient can undergo for the following parts of your body. How many X-rays does this equate to?
 - Head
 - Arm
 - Abdomen
 - Testes



For Better or For Worse

7a Disease Crossword



WORDS

- AIDS
- antibiotics
- antibodies
- antigen
- bacteria
- CJD
- disease
- fungi
- health
- HIV
- hot
- infectious
- malaria
- mental
- pathogens
- protozoan
- spiral
- virus

CLUES

Across

- 1 Procaryotic organisms, some of which can cause diseases. (9)
- 6 Any substance that can trigger an immune response in the body. (7)
- 7 Disease which uses a vector in its life cycle. (7)
- 10 Material produced by the body to help identify and combat foreign bodies or micro-organisms. (10)
- 12 Stands for Creutzfeldt-Jakob Disease. (3)
- 14 Our body feels this when we are fighting an infection. (3)
- 15 Acquired Immune Deficiency Syndrome caused by HIV (initials). (4)
- 17 Athlete's foot is an example of a disease caused by this type of organism. (6)
- 18 Bacteria are commonly found in three structures, of which this is one. (6)

Down

- 2 Substance naturally produced by some micro-organisms. It can also be made (synthesised) to help combat some diseases. (11)
- 3 Type of disease which can be spread from one person to another. (10)
- 4 The health of the mind. (6)
- 5 Disease-causing organisms are given this general term. (9)
- 8 Difficult to classify as living or non-living, has a protein coating and contains either RNA or DNA. (5)
- 9 A state of physical, mental and social wellbeing. (6)
- 11 Subkingdom of single-celled organisms. (9)
- 13 Interferes with the normal functioning of the body. (7)
- 16 Initials of the virus which causes AIDS. (3)

7b Health and Disease

To be **healthy** is to be in a state of physical, mental and social wellbeing.

Disease is any altering of this state of wellbeing and involves the interfering of the normal functioning of the body. Some examples of disease include the flu, common cold, mumps, measles, chicken pox, cancer, Down syndrome, rickets, hepatitis, meningitis, AIDS. Diseases can be infectious or non-infectious.

Infectious diseases are diseases that are said to be contagious, ie they can be transmitted from one person to another via air, water or contact.

- 1
- a

List as many infectious diseases that you can think of.

b

Describe an incident when you or a member of your family had an infectious disease.

Non-infectious diseases are diseases that interfere with the normal functioning of the body, however they are not contagious and can be caused by a variety of means, for instance:

- Nutritional diseases are caused by a lack of vital nutrients, eg vitamins or minerals.
- Genetic disorders are due to the inheritance of genes that interfere with the normal functioning of the body, eg Down syndrome.
- Environmental diseases occur where a chemical or something in our environment causes a disease, eg mercury poisoning or skin cancer due to exposure to the sun.

- 2
- a

List as many non-infectious diseases as you can think of.

b

If you know someone who suffers from a non-infectious disease, describe what limitations are placed on their life.

Pathogens are disease-causing organisms. They can be macroscopic or microscopic. Pathogens can be classified as: bacteria, fungi, protozoa, prion, viruses or macro parasites.

- 3
- In small groups, research and discuss the following diseases. Classify each disease and place it in the correct column in the table below.

Dysentery, Athlete's foot, Fatal Familial Insomnia (FFI), Ebola, Influenza, Mites, Ticks, Typhoid, Cholera, Thrush, Malaria, Mumps, Amoebic dysentery, Giardia, Pneumonia, Ringworm, Lice, Creutzfeldt-Jakob Disease (CJD), African sleeping sickness, Syphilis, Malaria, Bovine spongiform encephalitis (BSE), Chicken pox, Rubella, Smallpox, Whooping cough, Glandular fever

- 4
- An environmental disease that is prevalent in today's society is lung cancer. Discuss the possible causes of lung cancer. Evaluate and explain possible ways in which the number of cases of lung cancer could be reduced.

Bacteria	Fungi	Protozoan	Prion	Viruses	Macro parasites

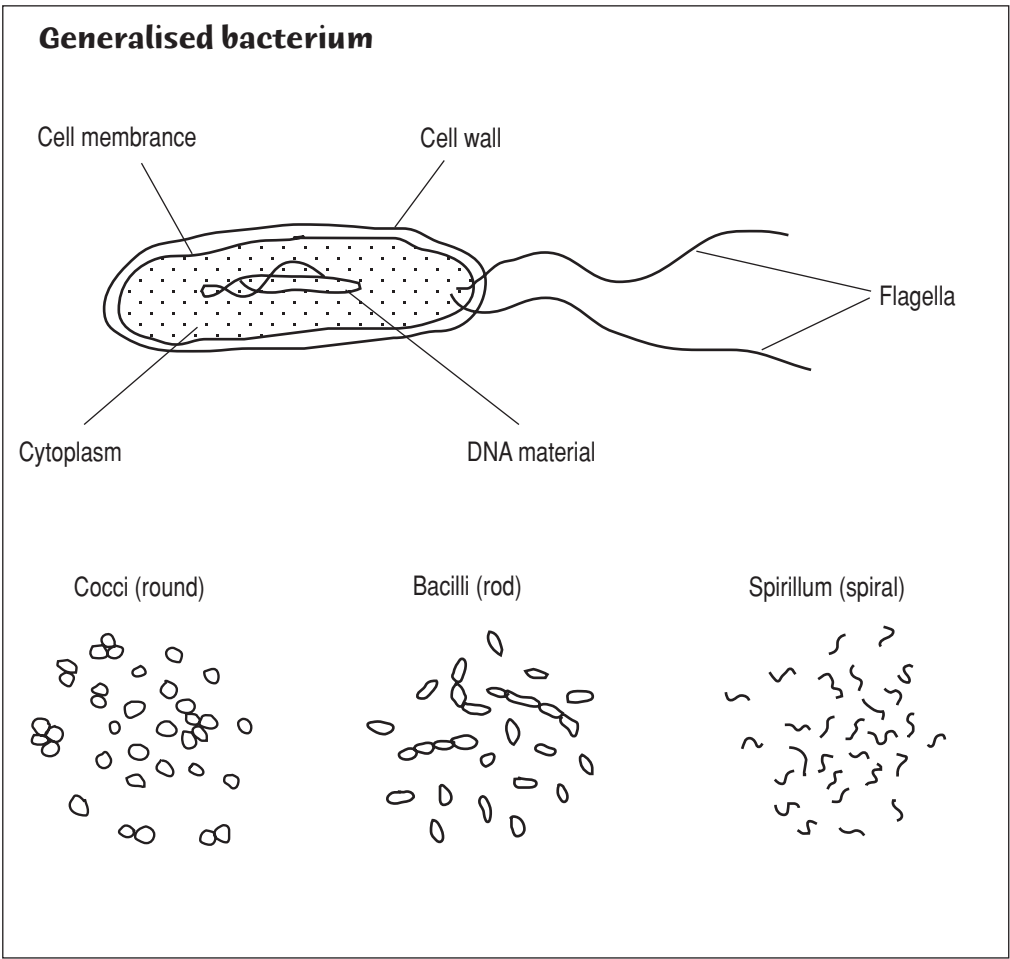
For Better or For Worse

7c Bacteria

Bacteria are unicellular organisms that are procaryotic (they do not contain membrane bound organelles). Bacteria can range in size from 0.5 to 5 μm . (1 μm is 1/1000th of a mm or 1 millionth of a metre).

Bacteria are classified according to their shape as shown in the figures below:

Cocci, or round-shaped bacteria, are often arranged in pairs, chains or clusters. These are called diplococci (paired), streptococci (in chains) or staphylococci (clusters). With Bacilli the cells can appear singularly or in chains, depending on the type of bacteria. They can also have flagella. The spirillum are given two names, Vibrio or Spirilla, depending on the bacteria.



1 Research activity: Use the library or internet to research the following bacterial diseases and complete the table below.

2 *Staphylococcus aureus* is commonly known as ‘golden staph’ and causes skin infections. Research this type of bacteria and describe how you could treat such an infection

Disease	Pathogen	Bacterial shape
a Cholera		Vibrio (spiral)
b	<i>Clostridium tetani</i>	
c	<i>Bordetella pertussis</i>	
d Syphilis		Spirilla (spiral)
e Tuberculosis		Bacilli (single cells)
f Pneumonia		Cocci (paired)
g Typhoid		Bacilli (with flagella)

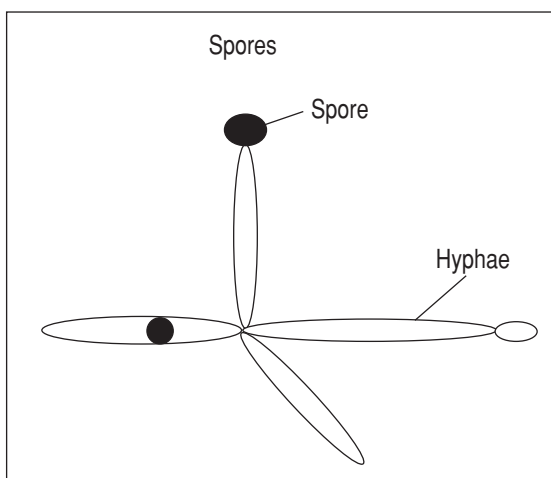
For Better or For Worse

7d Fungi

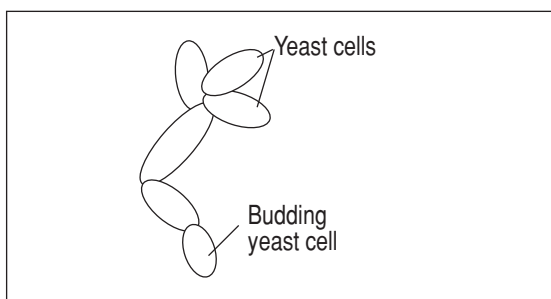
When we think of fungi we often think of the common mushroom. But, alas, there are much less tasty forms of fungi! Ever had that itchy feeling between the toes after showering in public showers? You may have had athlete's foot, or tinea. Ever found an itchy, scaly mark on your skin that looks like a small circle? You may have had ringworm. Both these diseases are caused by fungi, but are also very easily cured in our society.

Fungi are generally decomposers that break down just about any organic compound. Most, however, are parasitic and will break down the organic matter that is contained on living organisms, both plants and animals. The fungal cells secrete enzymes (organic catalysts) that promote digestion outside the cell, in this way large organic molecules are broken down into smaller molecules that the fungal cells can absorb and use. The fungi can reproduce using two main methods; sexual reproduction with the use of spores being produced by the hyphae, or, more commonly, via asexual budding or binary fission. Both of these methods are simplified in the diagrams below.

Generalised fungi



Yeast budding



- 1 Describe and define the following terms:
 - a binary fission
 - b spores
 - c enzymes

Practical activity

Candida albicans is the scientific name for yeast. Yeast is an example of a fungus that can easily be seen dividing. We call this budding. You are probably most familiar with yeast as being the organism that is used to help bread to rise or to brew beer. It can also cause a disease called thrush. You can observe yeast under the microscope. Take some dried yeast (you can buy this at the supermarket) and place about one teaspoon in half a cup (approximately 100 mL) of 10% glucose or sugar solution. Allow this to sit for about 5 minutes and observe what happens.

- 2 Record your observations by completing the following sentences.
 - a As soon as the yeast and 10% glucose solution were mixed I saw and heard and smelled ...
 - b After the yeast and solution had sat for 5 minutes I saw and heard and smelled ...
 - c 30 minutes after mixing I saw and heard and smelled ...

Now place a drop of the yeast solution onto a slide with a coverslip and observe under a microscope. You should see some budding.

- 3
 - a Draw a diagram of what you see, ensure you note the magnification.
 - b Add a written description of what happened to support your diagram.

For Better or For Worse

7e Protozoans and Vectors – Malaria

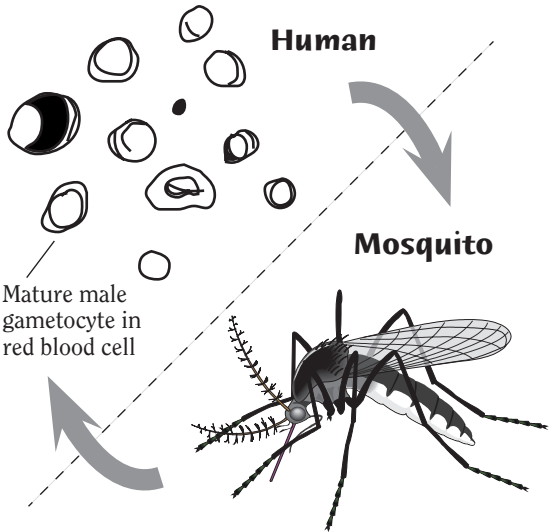
Protozoans are unicellular organisms that are parasitic in nature and were originally thought to be the ancient relatives to multicellular animals, in fact protozoan means ‘first animals’. Less than two dozen species of protozoans cause disease in humans.

There are many different types of protozoans; some have flagella, a tail-like structure that allows for movement, others have cilia, which are small hair-like projections. The third main type is called amoeba, which is a unicellular organism that physically engulfs its food.

- 1
- Investigate different types of protozoans and, in the spaces provided, draw three different examples. Label your diagrams showing the structural characteristics of each organism.

- 3
- Research malaria and rank the events listed below in the correct order in the life cycle of Plasmodium

LIFE CYCLE OF PLASMODIUM



Place the events listed below in the correct order in the life cycle of **Plasmodium**.

Rank Event

- ☐
- a
- Anopheles mosquito sucks up human blood and gametes.
- ☐
- b
- Cyst of fertilised zygotes moves to salivary glands of mosquito.
- ☐
- c
- Plasmodium forms gametes in gut of mosquito.
- ☐
- d
- Anopheles mosquito injects sporozoites into the human.
- ☐
- e
- Cyst of fertilised zygotes forms in stomach of mosquito.

--	--	--

- 2
- Define the following words:
a Cilia b Flagella c Amoeba

Malaria is an example of a disease that is caused by a protozoan called *Plasmodium falciparum*. It infects humans using a vector, an intermediary organism. In the case of malaria the vector is an anopheline mosquito. Malaria is most commonly found in warm climates such as Africa and Asia.

- 4
- Investigate malaria and use the following headings in a table to summarise this disease:

Causative organism, Vector, Host symptoms, Host response, Treatment, Control.

For Better or For Worse

7f Viruses and Prions

Viruses differ from bacteria and fungi and any other cellular organism. They contain a nucleic acid, either DNA (deoxyribonucleic acid) or RNA (ribonucleic acid), which is surrounded by a protein coat. Viruses range in size from approximately 30 to 300 nm. Viral diseases include influenza, chicken pox, small pox, AIDS and glandular fever. There are no cures for viral diseases at present, however, vaccinations do tend to reduce the chance of infection.

Viruses can be classified as:

Prions are infective proteins that cause the degeneration of brain tissue. The protein contains no genetic material to reproduce itself; instead the infective prion reproduces by taking over normal cellular prion protein in brain tissue and stimulating its conversion into the disease-causing prion.

(i) RNA viruses



Orthomyxovirus
e.g. influenza



Retroviruses
e.g. hepatitis, cancer



Paramyxoviruses
e.g. mumps, measles

(ii) DNA viruses



Herpes virus
e.g. chicken pox, shingles, cancer



Hepatitis B
e.g. hepatitis B



Pox viruses
e.g. small pox, cow pox

1 Describe how prions reproduce.

2 The table below contains a list of diseases caused by prions and viruses. Complete the table.

Type of pathogen	Disease caused	Pathogen	Host symptoms	Treatment or prevention
a Virus (DNA)	Small pox	Pox virus		
b Virus (RNA)	AIDS	Retrovirus		
c Virus (RNA)	Influenza	Orthomyxovirus		
d Virus (RNA)	Poliomyelitis	Poliovirus		
e Prion	Bovine spongiform encephalitis			
f Prion	Creutzfeldt-Jakob disease			

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For Better or For Worse

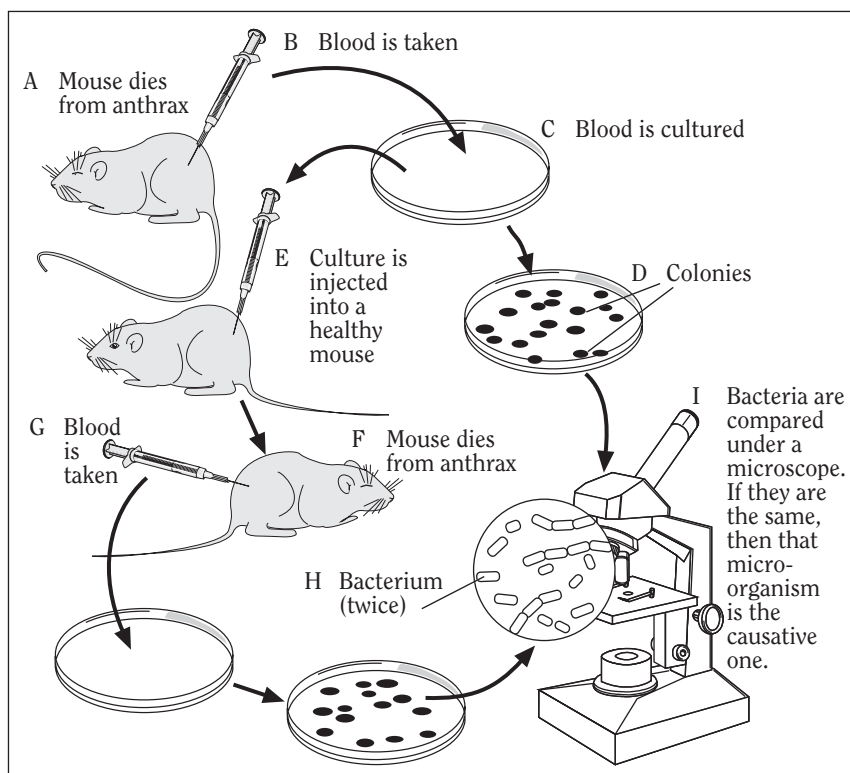
7g How We Identified Pathogens

ROBERT KOCH (1843-1910)

Robert Koch was a German physician. He developed a method of identifying the micro-organism that caused certain diseases. This method has been called Koch's postulates.

Koch's postulates

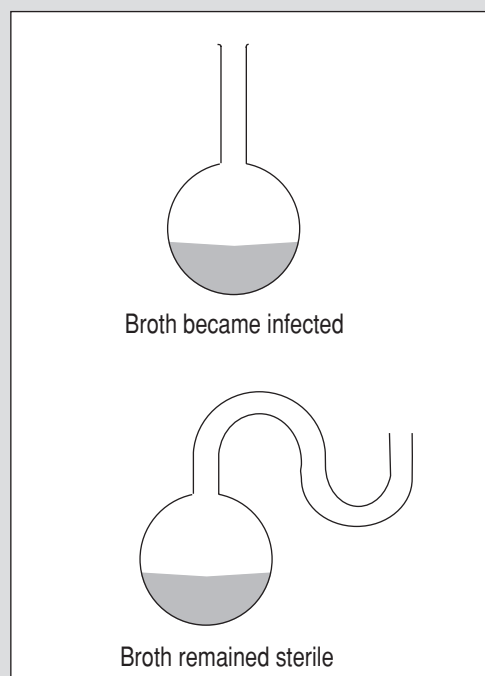
- The micro-organism that caused the disease must be present in all organisms suffering the disease.
- The micro-organism that caused the disease must be isolated in a pure culture.
- The micro-organism is injected into another organism (an experimental organism) and produces the same symptoms as the initially infected organism.
- The same micro-organism must be consistently isolated in pure culture from the experimental organism.



- 1 Imagine you have to SMS these postulates to a friend. How would you write them?
- 2 A new disease has just been found. Using the above information as a guide, describe how you would determine what organism caused the disease.

LOUIS PASTEUR (1825-1895)

Pasteur is remembered for disproving the theory of spontaneous generation. He did this by placing some broth in flasks as illustrated in the diagrams below. He heated this broth so that any organisms present were killed and the broth became sterile. He left the flasks and observed what happened. He noticed that the swan-necked flask remained sterile, while the other flask became infected and micro-organisms grew.



- 3 Describe why heating the broth was essential to this experiment.
- 4 In science we learn about a concept called an 'experimental control'. What was the control in this experiment? How did it help Pasteur?
- 5 Pasteur is responsible for the method of sterilisation we now call 'pasteurisation'. What is pasteurisation?
- 6 Pasteurisation is used to keep milk for longer. Describe how this works.
- 7 Pasteur also worked on developing a vaccination for anthrax. Explain what anthrax is.

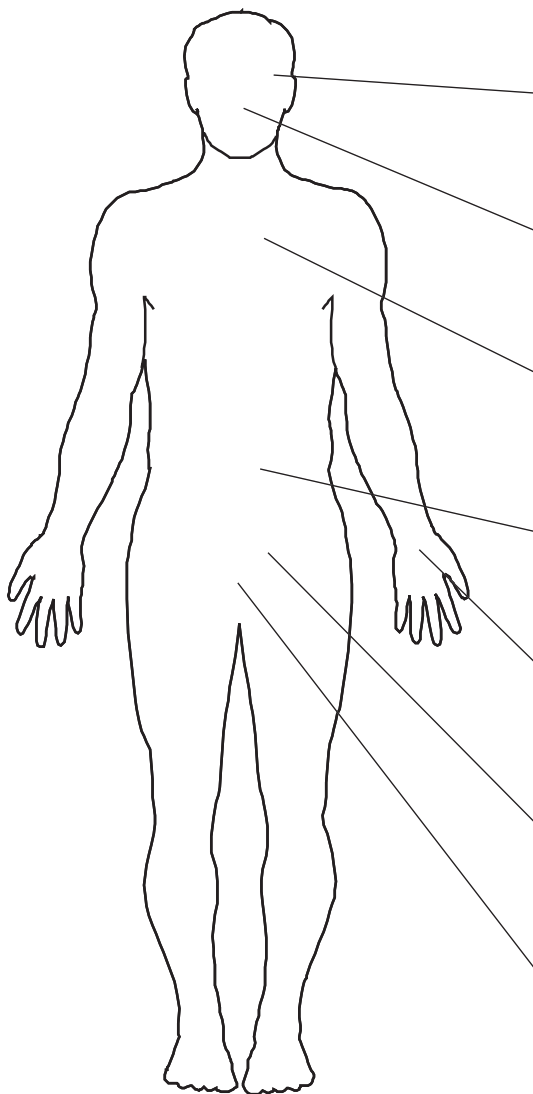
For Better or For Worse

7h The Body's Natural Defence

Diseases are often passed by water, air or through direct contact.

1 Describe some common practices that could be used to prevent contracting a disease.

2 The human body has many natural defence mechanisms. Explain how each of the labelled parts defends the body from pathogens.



Eyes:
Nose:
Respiratory tract:
Stomach:
Skin:
Intestine:
Urinary tract:

- 3**
- a** Name one body opening through which a pathogen could enter the body.
 - b** For this opening describe the natural defence that reduces the ability for the pathogen to take hold.
 - c** List some human behaviours which support our defence mechanisms against disease.
 - d** Describe some behaviours which are classed as risky and go against the body protecting itself.

- 4**
- a** Explain the importance of hand washing after using the toilet.
 - b** Name one disease that we could contract through unhygienic practices.

- 5**
- The body produces some substances that inhibit (prevent) disease-causing organisms. List the areas of the body that produce these substances.

7i Vaccines and Immunisation

During our lifetime we build up defence mechanisms. This is described as **acquired immunity**. One type, passive acquired immunity, involves the injection of antibodies that have been produced by another organism to help combat the disease. The other type, active acquired immunity, involves the direct injection of the antigen.

Vaccines are viral substances that have been treated to reduce the virulence of the virus, but are strong enough to allow for an immune response to be triggered. The vaccination prevents the disease because it initiates a response and allows the immune system to produce memory cells. This reduces time needed to combat the disease should it recur.

- 1
- Explain the following terms:
- a

Passive acquired immunity
- b

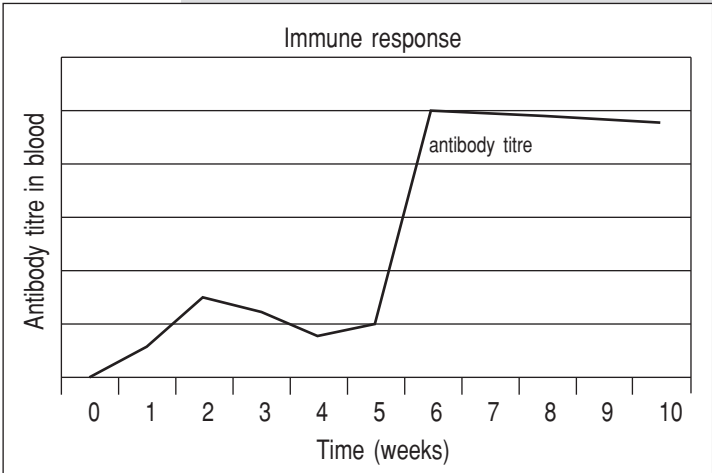
Active acquired immunity
- c

Vaccination
- d

Immunisation
- e

Vaccine

The graph below shows the immune response to a virus. At 0 weeks the patient is injected with a vaccine and at week 5 the patient is introduced to the antigen again.



- 2
- a

What is meant by the term antibody titre?
- b

Describe the response in the patient after the initial injection.
- c

Describe the response in the patient after the second injection.
- d

What is the advantage of having a higher antibody titre?
- e

When is the patient (described in the graph) at least risk to develop the disease?

3

a

Describe how vaccination prevents disease.

b

What is the difference between active and passive acquired immunity?

4

The SARS virus has caused serious health concerns in some parts of Asia. Describe what sort of practices could be introduced to limit the number of infections.

5

Smallpox appears to have been eradicated from the planet. No cases have been reported for a very long time. A vaccination program was put into place to help to wipe out this disease.

a

Investigate smallpox and describe the effect on a person infected by this disease.

b

What is the causative organism of smallpox?

c

Construct a timeline that shows the development of the smallpox vaccine.

d

If a person was immunised against smallpox, explain what the body's response would be. Describe this in terms of the immune response.

6

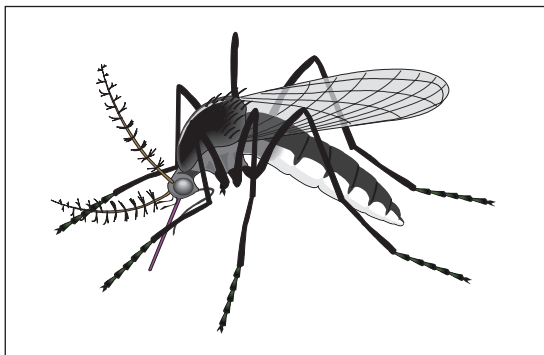
WHO stands for the World Health Organization. This group is responsible for putting into place vaccination programs for diseases that become prevalent in a society. Investigate WHO and write a discussion on two diseases that have been controlled by WHO programs.

For Better or For Worse

7j Research Project

A MALARIA

Malaria is a disease that is caused by a protozoan called *Plasmodium falciparum*. It infects humans using a vector of the female anopheles mosquito. Malaria is most commonly found in warm climates such as Africa and Asia.



This disease was first recorded as early as 18 BCE, but it has not been until recent times that the full cycle of the disease had been recognised. Research the developments that have occurred regarding knowledge

and treatment of malaria and construct a timeline to document these developments. Your timeline can begin at 1800 and go to present day.

The website <http://www-micro.msb.le.ac.uk/224/malaria.html> from the University of Leicester may be useful.

B ORGAN TRANSPLANTS

Sometimes an organ in the body does not function properly. This can be due to a number of reasons, such as a poor diet, disease or high-risk behaviours (eg smoking or drinking). At times organ transplants are used to extend the life of the patient. However, to be able to have an organ transplant the body's own immune system needs to be suppressed. Explain why this is so. Also explain how the immune system is suppressed.

The website http://www.med.monash.edu.au/medicine/alfred/research/organ_trans/ from Monash University may be helpful.

C VACCINATIONS

There is an ongoing debate as to whether people should be vaccinated against a disease or not, particularly in regard to young children.

You are to form two groups in which you will present a debate. Each group will argue one of the following points:

- *That vaccinations are vital and that each person should be vaccinated against diseases that are prevalent.*
- *That vaccinations are not vital and your body is able to defend itself.*

The website <http://www.avn.org.au/> from Australian Vaccination Network may be helpful.

D SMALL POX

Small pox was once a huge problem in the world. It is now rarely, if ever, seen or heard of.

Research the history of small pox and construct a timeline that describes the programs that were implemented to eliminate small pox.

The website http://www.health.gov.au/pubhlth/strateg/communic/factsheets/smallpox_fact.htm from Australian Department of Health and Aging may be helpful.

E ETHICS

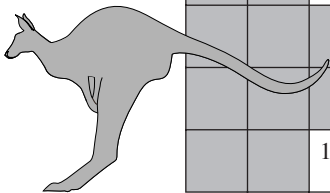
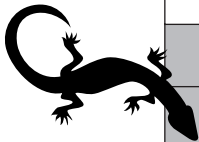
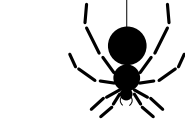
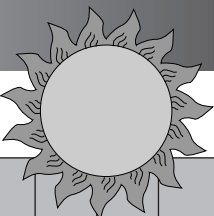
A couple has a child who was born with a rare genetic disease called hyper IgM syndrome. The couple have decided to have another child, however, because their first child has this disease, they are having their second child through IVF to screen for the disease. The second child is also being genetically matched with the first child so that a life-saving bone marrow transplant can be conducted.

DEBATE

Carry out a debate on whether you agree or disagree with this ethical issues raised in this situation.

Australian Ecosystems

8a Ecosystem Crossword



	1											2
3						4				5		
								6				
									7			
8												
		9							10			
11												
					12							
				13								
							14					
		15										

WORDS

- abiotic
- abundance
- biotic
- community
- distribution
- ecosystem
- fauna
- flora
- food web
- organism
- quadrat
- rot
- sun
- transect
- water

CLUES

Across

- 3 Term used to describe a large number of an organism present in an ecosystem. (9)
- 7 All the energy for the Earth comes from this place. (3)
- 8 Includes all living aspects of an ecosystem. (6)
- 10 To be recycled, material first has to decompose or _____. (3)
- 11 A line belt through an ecosystem that represents the distribution of plants. (8)
- 13 A small area used to sample an entire ecosystem, can be used to determine the abundance of an organism found in an ecosystem. (7)
- 14 Used to show all possible food chains in an ecosystem. (4,3)
- 15 Term used to describe all plants. (5)

Down

- 1 Includes all non-living aspects of an ecosystem. (7)
- 2 Word used to describe an individual plant or animal. (8)
- 4 Describes the animals inhabiting an area interacting with each other. (9)
- 5 An area of ecological study, includes all living and non-living components. (9)
- 6 The location of organisms within an ecosystem. (12)
- 9 Availability of this precious resource is essential for a healthy ecosystem. (5)
- 12 Term used to describe all animals. (5)

Australian Ecosystems

8b Studying an Ecosystem

An ecosystem is defined as the relationship between the physical environment and the biological environment (plants and animals). The physical environment involves such things as temperature, humidity, water availability, light etc. These are termed the abiotic factors. The biological environment comprises all the species of plants and animals that can be found in that ecosystem and are termed the biotic factors.

Discussion

- 1 In pairs, discuss possible reasons why it would be important to understand how to study an ecosystem. List each reason and write a summary of your discussion.

Ecosystems that are studied can range in size from a small pond to a large forest. Two examples of ecosystems are the Blue Mountains region of NSW and the Southern Coast of Victoria. To have a better understanding of an ecosystem it is essential that we study and measure both the abiotic and the biotic factors of the environment. Once these have been measured then it is possible to make correlations between the different factors. For instance, it may become evident that in areas where the soil has a slightly acidic pH the abundance of a specific type of plant may be more prevalent.



The Three Sisters (Blue Mountains, NSW)



Twelve Apostles (Port Campbell, Victoria)

- 2 What factors would you have to take into account if you were to study each of the following ecosystems?
 - a The Blue Mountains region
 - b The Southern Coast of Victoria

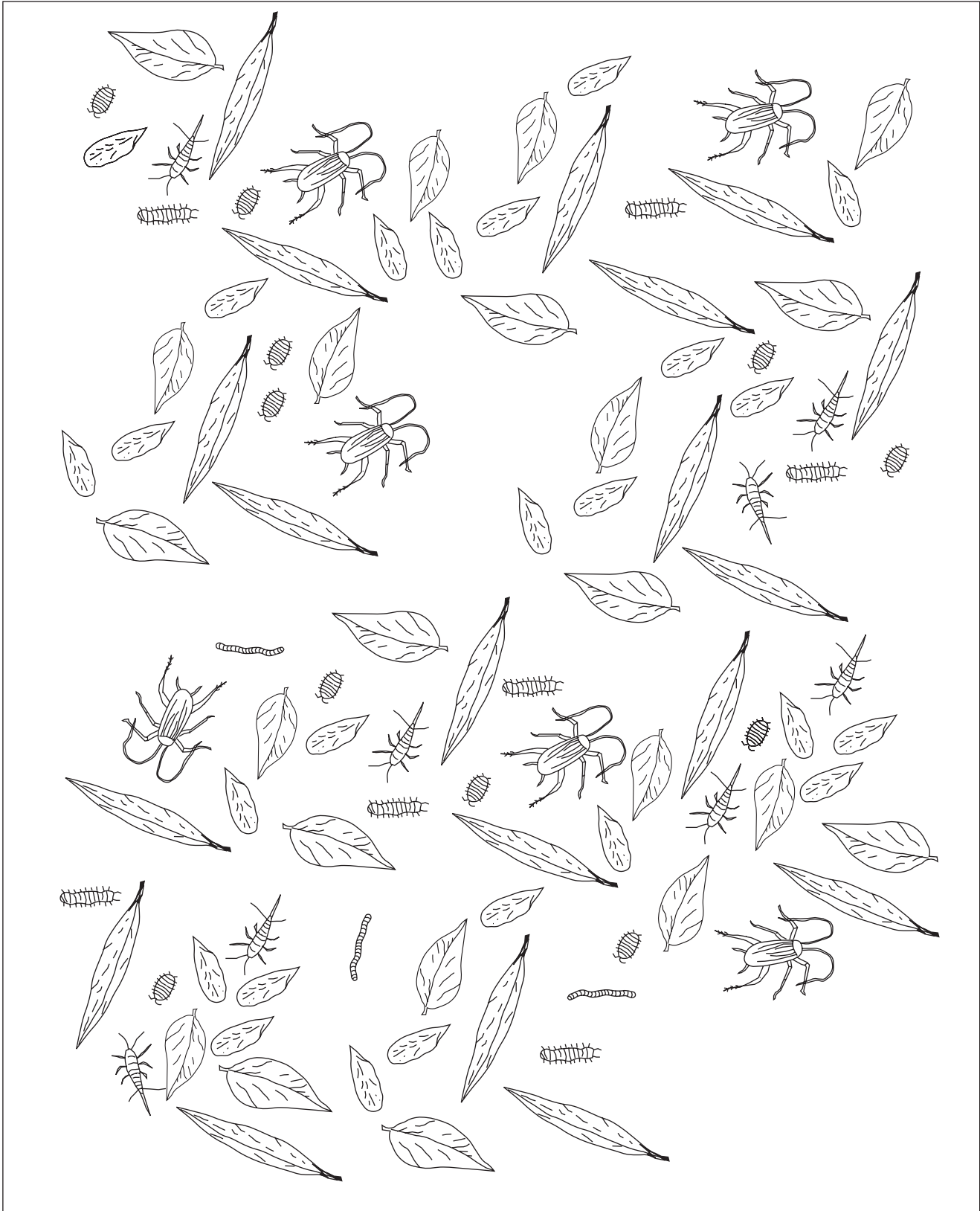
We can also make correlations between two biotic factors, for instance, if the number of lettuce plants decreases in your vegetable garden there may also be a decrease in the number of snails present. This sort of correlation can be represented as food chains and food webs.

- 3 Define the terms:
 - a biotic
 - b abiotic
 - c food chain
 - d food web
- 4 Draw a food web using any plants and animals (include insects, birds, reptiles and mammals) that you have noticed in your garden or local area.

Australian Ecosystems

8c Quadrat Sampling - A Soil Ecosystem

Below is an example of some leaf litter and the organisms that maybe present. You will need to use this ecosystem to perform the following quadrat sampling exercise.



Australian Ecosystems

8c Quadrat Sampling - A Soil Ecosystem (cont)

Quadrat sampling is a method used by biologists to estimate the number of organisms present in an ecosystem. This method saves time because a sample is taken to represent a whole area, preventing the whole area from having to be counted for each specific organism. This also saves money due to the fact that fewer resources and hours of labour are needed.

Using the leaf litter ecosystem that is found on the previous page, you will be able to conduct a study to determine the total number of each organism using the quadrat sampling method.

What to do

- 1 Cut a piece of cellophane (3 cm × 3 cm); this will be your quadrat.
- 2 You must drop a pen-tip onto your copy of the leaf litter ecosystem so that it leaves a small mark. This will represent the bottom left-hand corner of your quadrat. Arrange the cellophane so that the bottom left-hand corner is touching the mark and count the number of organisms in the quadrat. If you have an organism that is only partly in the quadrat, only count it if more than half of the organism is covered by the quadrat.
- 3 You must continue this process so that 10 quadrats are selected. For each quadrat use the results table below to tally up the results. Then calculate the total for each type of organism.

Results table

Quadrat Number	Organisms				
	Cockroach 	Millipede 	Silverfish 	Slater 	Worm 
1					
2					
3					
4					
5					
6					
7					
8					
9					
10					
TOTAL					

Australian Ecosystems

8c Quadrat Sampling - A Soil Ecosystem (cont)

Calculations

You must now calculate the number of organisms present in the whole area. Remember, so far you have only counted for a sample area. You need to use the following calculation:

Number of organisms in
the whole ecosystem

=

Area of the whole ecosystem

Area of the sample

×

Number of organisms
found in the sample

Example of calculation:

A soil ecosystem of 341 cm² was sampled for the slaters. 2 cm² quadrats were used and 10 quadrats were counted, the total number of slaters found was 47 slaters.

Number of slaters in the whole ecosystem

=

341/20 × 47

=

801.35

=

801 slaters in the entire soil ecosystem

Try this one!

An ecosystem measured 400 m² and a sample for *Banksia integrifolia* was conducted. Quadrats of 10 m² were used. The results are below. Calculate the total number of *Banksia integrifolia* found in the entire area in the space provided.

Quadrat	No. of <i>Banksia integrifolia</i>
1	3
2	0
3	0
4	2
5	5
6	0
7	1
8	2
9	0
10	4
Total	

Experimental calculation

- 1

For the experiment you have just performed calculate the number of cockroaches, millipedes, silverfish, slaters and worms that are present in the whole ecosystem.
- 2

Now count the number of each type of organism in the leaf litter ecosystem and record.

Evaluation

- 3

Compare the time taken to perform the experiment to that of counting all of the organisms. Imagine doing this in a forest!
- 4

How accurate were your calculations in the sample compared to the actual count?
- 5

In this type of example you would expect the sample to be a little different. Suggest reasons for this.
- 6

Describe how this method of sampling could be useful in determining where urban development could go ahead.

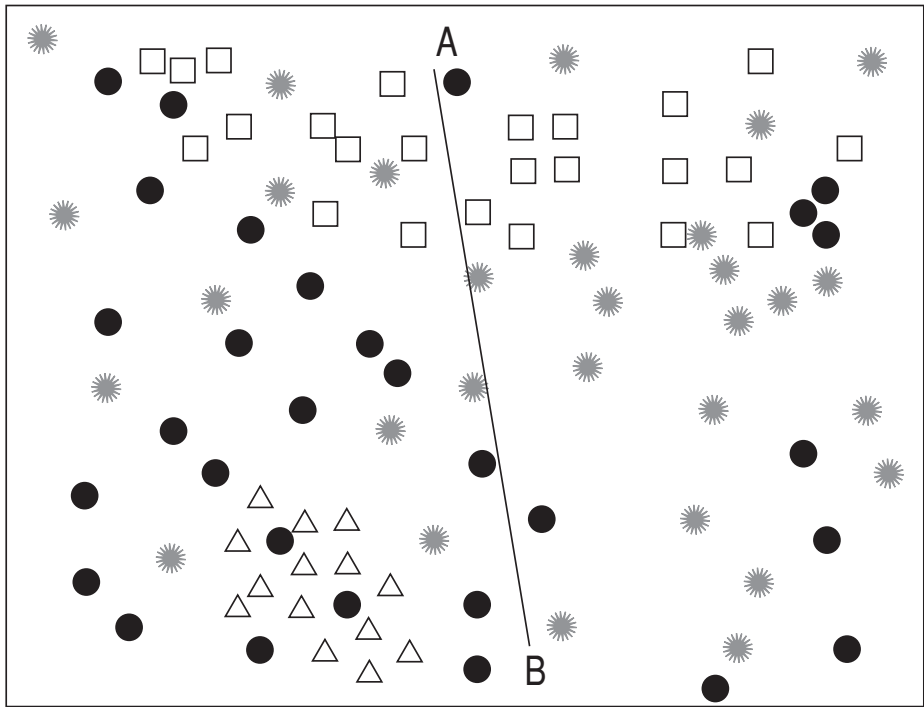
Australian Ecosystems

8d Distribution and Abundance – Transects

Distribution refers to where the organisms are located or spread out across an ecosystem.

Abundance is the number of organisms that make up that community.

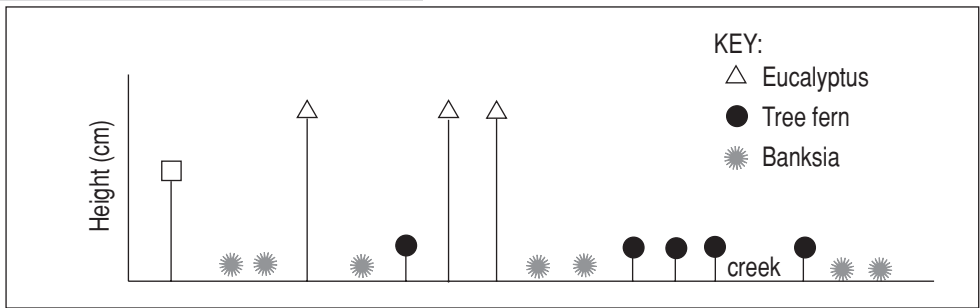
Several methods are used to determine the distribution and abundance of organisms without counting or mapping the location of each organism. These methods include: transects, quadrats, capture-recapture.



Line transects

A line transect can be formed by drawing a line from A to B on the above ecosystem. You would need to record on your line transect any plant that touches the line. (In the field this would be a tape). An example of a line transect is drawn on the diagram. A different line is used for the example below.

Sample line transect



- 1 What would be the advantages of a line transect in studying the environment?
- 2 What would be the disadvantages of a line transect in studying the environment?

Australian Ecosystems

8d Distribution and Abundance (cont)

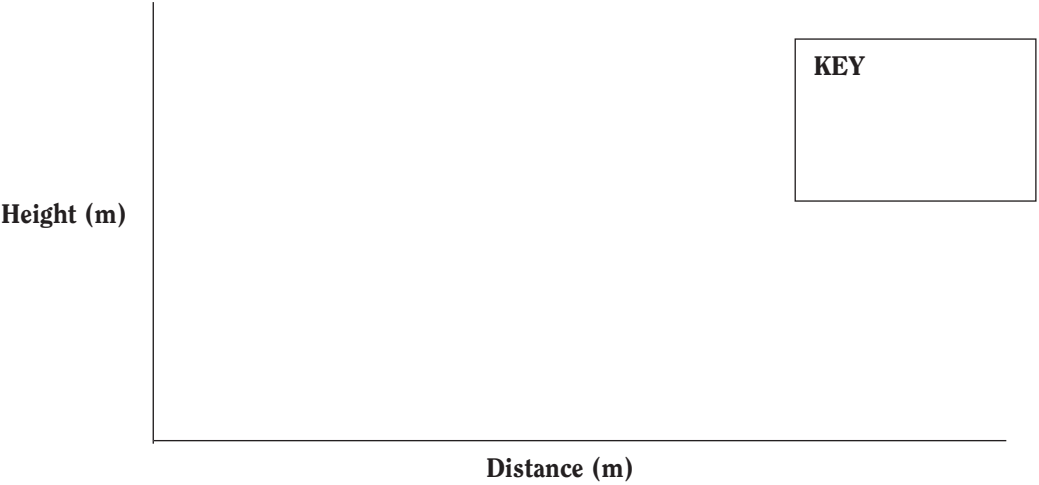
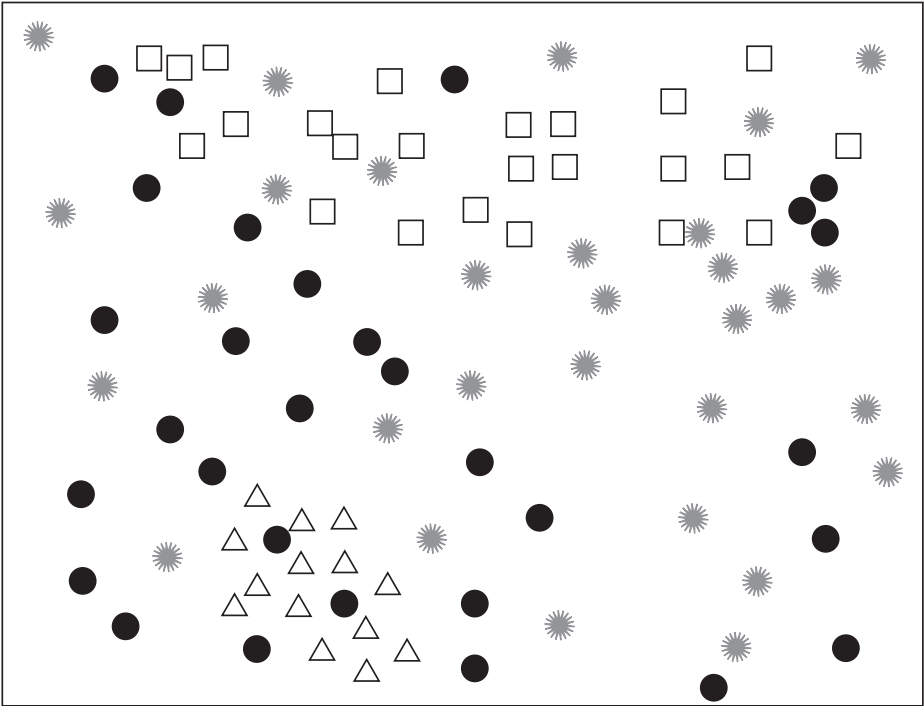
- 3

On the Distribution and Abundance grid (reprinted below) draw another line and construct another transect for this area.
- 4

Explain why it might be important for an ecologist to perform several transects of an area.
- 5

This sort of technique is used prior to many sorts of construction, for instance the construction of a new shopping centre. Discuss why this type of information may be useful to collect?
- 6

What factors could explain the distribution of one of the organisms being located in only one small area?



Australian Ecosystems

8e Biodiversity

Biodiversity describes the variety of living organisms within an environment. This variety of life includes different plants, animals and micro-organisms. The number of species or different types of living things in the world is not exactly known but there are thought to be between 10 and 30 million different species on Earth. Many organisms that we do know of are not well understood. We don't know how and where they live or how they interact with other species of organism or with us.

The number of different species in the world is being slowly decreased. This decreasing biodiversity is due to pollution and human intervention, such as hunting and logging. Native species continue to have their food eaten, or are killed themselves, by feral animals such as domestic cats and dogs that have escaped to the bush.

In Australia's history many animals and plants have become extinct since the first humans inhabited the continent. Currently about 25% of Australia's native plant species are under threat of extinction. In Sydney alone ten mammal species have become extinct since 1788. If a large number of species disappear at once then there could be more extensive disruption to the environment.

Most Australia's diverse wildlife is unique. About 80% of species occur only in Australia. This is because of the isolation of Australia over the last 150 to 200 million years. Recently introduced species are affecting these unique animals and plants. The most famous of these are the dingo (thousands of years ago) and, more recently, the rabbit and the cane toad.

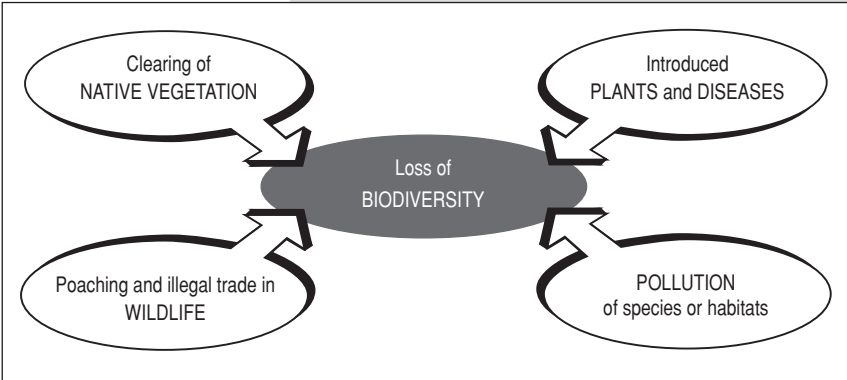
Humans have been changing the Australian environment for the last 60 000 years but in the last century the rate of change has become much greater. Since 1788 almost 90% of woodlands and 50% of rainforest has been cleared and destroyed. This change has allowed soil erosion to destroy farming areas as well as the edges of the forests that remain.

- 1** Describe, in your own words, the following terms:
 - a biodiversity
 - b environment
 - c micro-organism
 - d species
 - e pollution
 - f feral
- 2**
 - a What proportion of Australian plant species is thought to be under threat?
 - b How many mammal species have become extinct since 1788?
 - c Why do certain organisms become extinct?
 - d What could it mean for us if large numbers of species became extinct at the same time?
- 3** Suggest some reasons why we might be experiencing reduced biodiversity in our environment today.
- 4**
 - a What proportion of Australia's wildlife is unique to Australia?
 - b Why is this number so large?
- 5**
 - a List some introduced species of animals or plants.
 - b Choose one example of an introduced species and describe:
 - i) where it lives
 - ii) what it eats
 - iii) what problems are caused by this organism.
- 6** The Wollemi pine (*Wollemia nobilis*) is a conifer that was discovered in 1994. It is considered to be a living fossil as there is only a small number of these trees, located near Sydney, and because they are related to a genus of plant that dates back 150 million years ago. Several plans have been put into place to try and preserve this species of plant. Research the Wollemi pine and discuss the preservation projects that are in place.

Australian Ecosystems

8f Losing Our Biodiversity

Diversity of ecology is the range of different parts of the environment. An ecosystem is made up of the living communities of plants, animals and micro-organisms, as well as the non-living components such as soil, water and air, which are used as food and shelter for the living organisms. Some reasons for loss of biodiversity are shown in the diagram. Further explanation is given below.



- Clearing of native vegetation for either housing, forestry, mining or farmland encourages erosion and reduces the quality of the soil. In eastern NSW and Victoria over 90% of native vegetation has been removed.
- Introduced species can become competitors, predators or parasites on native wildlife. Many introduced micro-organisms cause diseases to which native wildlife has no natural immunity.
- Poaching and the illegal trade in native wildlife is the modern-day equivalent of hunting. This can cause the destruction of large numbers of the most spectacular and rare of native species, birds and reptiles in particular.
- Pollution of the environment can destroy native plants or make it easier for introduced species to thrive. Either way the original native organisms are placed under further pressure.

- 1 Give a description for each of these four reasons in your own words.
- 2
 - a Which of the four reasons for loss of biodiversity do you think is the greatest problem? Explain why.
 - b What can you or your friends do to reduce the problem caused by this particular reason?
- 3
 - a Can you see evidence of loss of biodiversity in your local area?
 - b If 'yes', describe what has caused the loss. If 'no' describe how your area keeps its biodiversity.
 - c Why it is important to maintain ecological biodiversity?

Research Project

Each local government or council has specific regulations concerning the zoning of the land in their particular area and the amount in which humans can impact upon this land. Research the area in which you live and list the different zonings in your area and what the criteria for each zoning is. The following categories should be included: domestic housing, units, schools, public transport facilities, commercial development, primary produce and industry.

For your particular area describe the sampling methods that could have been used to determine the boundaries for each of these zonings. In particular look at areas within your local environment that have been protected and find out why. The following websites may be helpful.

I.C.L.E.I.
<http://www.iclei.org/>

Sutherland Shire Council
<http://www.sutherland.nsw.gov.au/>

Landcare Australia
<http://www.landcareaustralia.com.au/about.asp>

City of Sydney
<http://www.cityofsydney.nsw.gov.au/>

Suggested Answers

5 ENERGY FOR OUR LIFE

5a Energy Crossword

Across: 5 hydro 7 sound 8 gas 9 heat 12 radiate 14 glucose
15 PE 16 cold 17 waste 19 joule 22 light 23 oil 24 solar 25 mgh
28 diesel 29 tide

Down: 1 jog 2 fossil 3 nonrenewable 4 energy 6 rot 10 atomic
11 renewable 13 alcohol 18 petrol 20 thermal 21 coal 24 sun
26 hot 27 KE

5b Sources of Energy

1 a Sun's energy causes wind to move which turns propeller moving a dynamo producing electricity. **b** No greenhouse gases and no other waste products. **c** Noise pollution and visual pollution.

2 a Fuel from fossilised remains of animals or plants. **b** Animal or plant remains are compressed under pressure for millions of years and only the carbon products remain. **c** It cannot be reproduced at the same rate at which it is used.

3 a They cannot be used for other purposes (such as plastics or drugs), they produce greenhouse gases when burned. **b** Easy to obtain, easy to transport, generally non-toxic to handle, release energy at accessible temperatures. **c** No because the electricity is produced by a turning dynamo regardless of the fuel used to turn it. **d** They don't release CO₂ or other waste gases.

4 a Yes because motors and generators get hot and produce sound and this heat and sound is lost energy. **b** Whenever there are moving parts in a machine, friction causes heat to be lost. This heat is energy which cannot be used for useful purposes. **c** Answers will vary.

5c Energy from the Wind

1 a wind, tidal and direct solar **b** Both use energy from the sun transferred through moving fluids on earth. Passive because it does not use the sun's energy directly, but uses its heating of the fluids on the earth. **c** Wind is variable and can occur for many hours at a time. Tidal is more regular and predictable but changes in a fixed cycle.

2 a, b & c During daytime, the warm air over the land rises and cool air from the sea takes its place resulting in a sea breeze. During night, the warm air over the sea rises and cool air from the land takes its place resulting in a land breeze. This difference is because water holds its temperature more than land.

3 a i) 13.8 turbines **ii)** 6.6 MW per farm **b** The number of homes that could be serviced by that amount of energy. **c** 302.8 **d** 0.0016 MW

4 a Good, because these gases pollute the atmosphere. **b** Energy which can be produced closer to the consumer.

5d Energy from the Ocean

1 a The water moves in and out of the system causing the turbine to be turned. **b** At the high and low tide points.

c Midway between high and low.

2 a Gravitation effects of sun and moon and the rotation of the Earth. **b** The sun's gravitation energy. **c** Renewable because it is being produced at a greater rate than we can use it.

3 a The wave pushes air up the chimney turning the turbine. **b** The moving turbine produces electricity by causing magnets to move past a coil of wire. **c** The wave creates a low pressure area in the chimney and atmospheric pressure pushes air down the chimney turning the turbine. **d** Electricity is produced by the turning turbine. It does not matter whether the turbine is turned by the air moving up or down the chimney.

4 a It does not produce waste chemicals or pollutants. **b** It is visually disturbing and could cause changes to the natural wave cycle.

5 a Wind caused by solar heating of the earth. **b** solar energy **c** Renewable because it is being produced at a greater rate than we can use it.

6 Answers will vary.

5e Electricity and Power

1 a *joule (J)*: a unit of energy; *watt (w)*: a unit of power; *power*: energy used per unit time; *megajoule*: 1 million joules; *megawatt*: 1 million watts **b** 800 **c** $60 \times 60 \times 60 = 216,000 \text{ J}$ **d** $1000 \times 60 \times 60 \times 8 = 28,800 \text{ kJ}$ **e** $2.5 \times 4 = 10 \text{ kWh}$

2 to 5 Answers will vary.

5f Solar Energy

1 a Answers will vary. **b** Include: low cost, overcomes difficulties in connecting to mains electricity, no pollution. **c** Start up costs. **d** There is not a need to set up wires over large distances for a small number of users.

2 a Electrons flow freely through a conductor. **b** Conductors include all metals, graphite, salt solutions. **c** Most non-metal materials. **d** There is a limited number of spaces where electrons can move therefore flow is reduced. **e** It conducts electricity, melts, conducts heat, solid at room temperature, forms ionic bonds. **f** Does not conduct electricity or heat. Solid (but many are not), forms co-valent bonds. **g** Solid, semiconductor of electricity, forms molecular bonds.

3 a solar energy **b** silicon **c** Positive proton, neutral neutron and much smaller negative electron.

4 Answers will vary.

Suggested Answers

5g Using Solar Energy

1 a Makes the solar cell perpendicular to the sun's rays - maximum coverage. **b** 0° **c** Nov and Dec at 30°; 29.9 MJ **d** Nov at 70°; 19.1 MJ

2 d Close to 30°. **e** The tilt angle should equal the latitude.

5h Chemical Energy

1 a The sun's energy contained when the plants or animals were alive. **b** They are used at a faster rate than they are produced. **c** Oil is used for transport, coal for electricity production and steel manufacture, gas for household energy.

2 a Oxygen and fuel. **b** CO₂ and H₂O **c** In incomplete combustion other chemicals are produced as waste. These could be used for further energy production. **d** Ash or any small carbon compound.

3 a black coal **b** transport **d** Black coal is a cleaner (less complex waste molecules) and therefore more efficient fuel.

4 a We will eventually run out of fuel. **b** Irritate the lungs leading the cancerous growths on the epithelial (skin) lining. **c** smoking **d** CO₂ production. **e** Increases in sea level and unusual weather patterns; melting of polar icecaps. **f** Answers will vary but include reduction in energy use and more efficient use of the energy we need.

5i Energy Consumption

1 a 60% **b** 24% **c** 1 pJ is 1 million MJ and 1 MJ is 1 million J. **d** 3500 PJ **e** 6000 PJ

2 a 31.1% and 221.1% **b** Vietnam is developing at a much faster rate. **c** Canada **d** It is a large country and therefore needs energy for transport; it is also very cold in winter and uses a lot of energy for heating (minus 30°C is a common winter temperature). **e** 1200 mill x 151 = 181,200 PJ for China and 11,627 PJ for Vietnam.

6 BIONICS

6a Bionics Crossword

Across: 1 ultrasound 3 aorta 4 xray 5 bicuspid 9 UHMWPE
11 heart 13 alveoli 14 trachea 15 hinge

Down: 2 thermograph 6 superalloy 7 diastolic 8 sinoatrial
10 keyhole 12 valve

6b Biomaterials

1 Contact lenses - eyes; pacemaker - heart; artificial hip; breast implants etc.

2 a Used for hip replacements as it is very strong, long lasting and biocompatible. **b** Smooth and biocompatible, used to coat the ball in a ball and socket joint. **c** Used for small parts eg end of finger or breast implants; biocompatible and flexible.

6c Heart

1 The left side of the heart is thicker than the right because it needs to pump blood all the way around the body, where the right side only needs to pump blood to the lungs.

2 To supply the heart with blood and oxygen.

3 A Bicuspid valve, B Left atrium, C Semilunar valves, D Left ventricle, E Right ventricle, F Vena cava, G Tricuspid valve, H Right atrium, I Sinoatrial, J Pulmonary artery, K Aorta, L Pulmonary vein.

4 *Aorta:* thick artery, vessel with very muscular walls ~1.5 - 2 cm diameter. *Left ventricle:* thick-walled heart chamber. *Right ventricle:* thinner than the left ventricle. *Left atrium:* small chamber of the heart. *Right atrium:* small chamber on the right side of the heart. *Sinoatrial node:* Found near the right atria. *Vena cava:* vein, joins heart at right atria. *Pulmonary artery:* carries blood away from the body. *Pulmonary vein:* carries blood to the heart from the lungs. *Bicuspid valve* and *Semilunar valves:* valves found in left ventricle. *Tricuspid valve:* valve found in right ventricle.

6d ECG - Electrocardiographs

1 If the heart has an irregular beat or patient has experienced chest pain.

3 a P wave - 5.75; Q wave - 5.82; R wave - 5.9; S wave - 5.92; T wave - 6.1 **b** P-R - 0.15; QRS - 0.1 s; Q-T - 0.28

5 The heart beats due to electrical impulses from the sinoatrial node. When this does not function correctly, the heart can be given another electrical shock to get it back into rhythm.

6e Pacemakers and Heart Valves

1 a Bicuspid valve is found in the left side of the heart between the atrium and the ventricle, it has two points or attachment areas. **b** The semilunar valve covers the opening of the pulmonary artery. **c** Tricuspid valve has three attachment points and is found in the right side of the heart.

2 Answers will vary but the conclusion is to keep the blood flowing in one direction.

3 Ball and cage valve: Made of a metal housing and carbon ball, which opens and closes allowing the blood to flow. Bi-leaflet valve: Consists of two carbon disks which open and close.

5 a *Arrhythmia:* abnormal heart beat. **b** *Bradycardia:* slowness of the heart beat. **c** *Tachycardia:* quickness of the heart beat. **d** *Diastolic:* pressure reading when the ventricle relaxes. **e** *Systolic:* pressure reading when ventricles contract and blood is pumped into the arteries.

6 Breathlessness, lethargy, tiredness, blue lips or fingertips.

Suggested Answers

6f Skeleton and Joints

3 • Ball and socket – superalloys and UHMWPE. • Silicone is only used in small joint areas where there is little weight-bearing force..

6g Minimal Invasive and Surgical Techniques

1 In normal surgery, the surgeon is able to fully manipulate the organs etc, which can increase pain. Minimally invasive surgery limits the stay in the hospital and reduces recovery time, however manipulation is limited to what instruments can do.

2 a The time spent in hospital is much shorter today than the 1970s. **b** The shorter stay in hospital may be because many operations are now performed by minimally invasive techniques whereas in the 1970s full surgical techniques were used. Technology and knowledge of treatment, infection etc has also improved.

3 a *laparoscope*: inspects the abdominal area. **b** *gastroscope*: inspects the gullet and stomach area. **c** *endoscope*: allows for visual inspection of inside the body. **d** *endoscopy*: technique used to visually inspect the internal organs. **e** *fiberscope*: device that contains optical fibres for viewing the internal organs.

4 a tubular ligation, appendectomy, gall bladder removal, exploratory surgery **b** Specialised instruments are inserted into very small incisions and the surgeon performs the operation by looking at a screen.

6 Due to large incision there is more chance of infection, more pain, longer recovery period, larger scar. Tiny incisions in keyhole surgery minimise all these factors.

6h Non-invasive Medical Techniques

1 *Advantages*: allows for diagnostic techniques to occur without entering the body. *Disadvantages*: the doctor must rely on the evidence from the machine. Because no manipulation can take place, assumptions may need to be made in the diagnosis.

2 a *X-ray*: to determine broken bones. **b** *Ultrasound*: imaging of internal organs using pressure waves; used to check foetus in the uterus. **c** *PET*: provides images of blood flow etc.

d *Thermography*: uses infrared images which can show increased blood flow to a certain area. **e** *MRI*: detailed images of body tissue, these images can be taken from any angle.

3 Answers will vary.

4 Non-invasive techniques require no entry into the body, whereas minimally invasive techniques do use some small penetration of the body.

5 Answers will vary.

6 a X-ray **b** CT scan. **c** ultrasound. **d** MRI **e** thermography

6h The Respiratory System and Respirators

2 a *Nasal cavity*: collects air. **b** *Alveoli*: sacs that allow oxygen to diffuse into the blood. **c** *Lungs*: major organs of the respiratory system. **d** *Intercostal muscles*: aid the opening and the closing of the abdominal area which allows the lungs to expand and contract. Assist with the diaphragm to enlarge the chest cavity. **e** *Bronchi*: takes air from the trachea to the lungs. **f** *Trachea*: wind pipe. **g** *Capillary network*: network of pipes that allow for oxygen to be diffused into the blood so that the cells can be supplied with oxygen for cellular respiration.

3 Answers will vary.

4 Any operation that requires the opening of the chest cavity would require an artificial lung. Once that area is opened there is no control of the pressure, ie expansion and contraction which allow for oxygen and air to enter the lungs with the aid of the diaphragm and the intercostal muscles.

FOR BETTER OR FOR WORSE

7a Disease Crossword

Across: 1 bacteria 6 antigen 7 malaria 10 antibodies 12 CJB 14 hot 15 AIDS 17 fungus 18 spiral

Down: 2 antibiotics 3 infectious 4 mental 5 pathogens 8 virus 9 health 11 protozoan 13 disease 16 HIV

7b Health and Disease

1 a Flu, cold, AIDS, hepatitis, meningitis, mumps, measles, chicken pox. **b** Answers will vary.

2 Down syndrome, scurvy, osteoporosis, cancer. **b** Answers will vary.

3 *Bacteria*: dysentery, cholera, typhoid, giardia, syphilis, pneumonia, whooping cough. *Fungi*: athlete's foot, thrush, ringworm. *Protozoan*: malaria, amoebic dysentery, African sleeping sickness. *Prion*: Fatal Familial Insomnia (FFI), Creutzfeldt-Jakob disease (CJD), bovine spongiform encephalitis (BSE). *Viruses*: ebola, influenza, mumps, chicken pox, rubella, smallpox, glandular fever. *Macroparasites*: mites, ticks, lice.

7c Bacteria

1 a Cholera, *Vibrio cholerae*, *Vibrio* (spiral); **b** Tetanus, *Clostridium tetani*, Rod; **c** Whooping cough, *Bordetella pertussis*, Cocci; **d** Syphilis, *Treponema pallidum*, *Spirilla* (spiral); **e** Tuberculosis, *Mycobacterium tuberculosis*, *Bacilli* (single cells); **f** Pneumonia, *Streptococcus pneumoniae*, Cocci (paired); **g** Typhoid, *Salmonella typhi*, *Bacilli* (with flagella).

7d Fungi

1 a *binary fission*: division of a cell or nucleus into two daughter cells. **b** *spores*: the reproductive gametes of fungi. **c** *enzymes*: biological catalysts.

Suggested Answers

2 Answers will vary but include noticing some bubbling of the solution, where CO₂ is formed.

3 Answers will vary.

7e Protozoans and Vectors - Malaria

2 **a** *Cilia*: small hair-like projection. **b** *Flagella*: tail-like structure that allows for movement. **c** *Amoeba*: type of protozoan, a unicellular organism that engulfs its food via phagocytosis.

3 **c, e, b, d, a.** A mosquito sucks blood from a human. If it is infected with the Plasmodium protozoan, then bites another human, the infected blood is injected into the person's bloodstream. The protozoan then moves to the person's liver where it multiplies and attacks the red blood cells, causing the symptoms of malaria.

4 **Causative organism**: Plasmodium vivax. **Vector**: Anopheles mosquito. **Host symptoms**: fever, chills, sweating. **Treatment and control**: can be controlled by taking quinine in high risk areas. Chloroquine phosphate and quinine sulfate are the substances of choice for treatment.

7f Viruses and Prions

1 Prion proteins can work on a molecular level similar to the brain tissue so therefore is able to make more prions by converting healthy cells.

2 **a** Virus (DNA), small pox, pox virus, rash, vaccination.

b Virus (RNA), AIDS, retrovirus, sores, nil, education and safe sex. **c** Virus (RNA), influenza, orthomyxovirus, coughing, fever, vaccination. **d** Virus (RNA), poliomyelitis, poliovirus, cold-like symptoms, diarrhoea, vomiting, fever, vaccination.

e Prion, bovine spongiform encephalitis, destroys brain tissue.

f Prion, Creutzfeldt-Jakob disease, neurodegenerative disease, dementia, muscle spasms, blindness, suspected carriers should not donate blood.

7g How We Identified Pathogens

1 Answers will vary.

2 Answers will vary but must include all of the stages of Koch's postulates.

3 By heating the broth you are making it sterile.

4 The control is the normal broth - unaffected by the experiment.

5 Pasteurisation is the process of heating a substance to remove the micro-organisms.

6 The milk is heated so that it kills the micro-organisms that will spoil the milk, however you do not heat the milk enough to destroy the protein structure of the milk.

7 Anthrax is an infectious disease caused by Bacillus anthracis. It causes pustules at the site of infection, respiratory distress, coma and can cause death.

7h The Body's Natural Defence

1 Cover your mouth when you cough or sneeze. Do not share drinking glasses or cutlery. Cook food thoroughly. Wash hands. Keep clothes, towels, sheets etc clean.

2 **Eyes**: antimicrobial substances in the eye fluid. **Nose**: mucus to collect any pathogens. **Respiratory system**: cilia to catch pathogens and mucus to collect them in. **Stomach**: acidic conditions that will kill micro-organisms. **Skin**: dry barren environment that inhibits micro-organism, sweat has antimicrobial substances in it. **Intestine**: alkaline condition and after stomach (highly acidic) inhibits bacterial pathogens, also constant movement and flushing of system inhibits disease. **Urinary tract**: constant flushing and antimicrobial substances.

3 **a** eg mouth, nose, ears **b** Answers will vary, eg pathogens entering the mouth through food are often killed by the acidic environment in the stomach. **c** Washing hands and food; covering mouth and nose when coughing or sneezing; taking extra vitamins; avoiding close contact with contagious people etc. **d** Being unhygienic, food, unsafe sex etc.

4 **a** Removal of dirt and surface micro-organisms from hands. Reduction in the number of pathogens which enter our body

b We could catch any of a number of diseases from food poisoning to cholera or diarrhoea.

5 eg eyes, stomach, urinary tract

7i Vaccines and Immunisation

1 **a** The injection of antibodies that have been produced by another organism to help combat the disease. **b** The injection of the antigen (a weakened version is often injected). **c** Production of active immunity against a disease. **d** The administration of an antigen to produce an immune response. **e** Substance that has been treated to reduce virulence, allows for immune response to be triggered.

2 **a** Volume of antibody in blood. **b** Low level of resistance.

c Much higher level of antibody. **d** Much better response to disease. **e** After the second injection.

3 **a** Increasing the body's recognition of the micro-organism and its ability to fight it. **b** Active and passive acquired immunity are different because in active the pathogen is injected into the body whereas in passive, some antibodies are injected from another organism.

4 If a vaccine could be made it could help other people from contracting the disease. Quarantine any person that is suspected of having the disease. Taking hygiene to a higher level, covering your mouth, constant cleanliness.

Suggested Answers

AUSTRALIAN ECOSYSTEMS

8a Ecosystem Crossword

Across: 3 abundance 7 sun 8 biotic 10 rot 11 transect

13 quadrat 14 food web 15 flora

Down: 1 abiotic 2 organism 4 community 5 ecosystem

6 distribution 9 water 12 fauna

8b Studying an Ecosystem

1 Answers will vary, such points could include:

To limit the damage to the local environment.

To determine if endangered species are present in the area being studied.

To help select certain areas, could be used to determine which area has the largest trees therefore more difficult to remove if construction is to take place.

2 Answers will vary, could include the size of the area to be studied; the effect of the season and weather on the area; access to the area etc.

3 **a** All living material. **b** All non-living material. **c** The relationship of organisms, their food and those who eat them.

d The relationship of all organisms in the environment.

8c Quadrat Sampling - A Soil Ecosystem

Try this one: Total number of organisms = 17 *Banksia integrifolia*

Number of organisms in the whole ecosystem = 68 *Banksia integrifolia*

Evaluation:

3 The time taken to count would be considerably greater than that of performing the calculations.

4 Answers could vary, the greater the size of the area and if the number of organisms are relatively spread evenly would allow for quite an accurate estimation of the actual count, however, if the sample of organisms is only confined to one area than there may be some error.

5 You would expect the sample to be different because it covers such a small area.

6 It would be useful because you could determine which areas could be built on that would have the least impact on the local environment.

8d Distribution and Abundance - Transects

1 *Advantages:* quicker than counting all of the organisms in area, gives a general idea of what organisms are present. Can give an overview of the whole area as lines across the whole area can be drawn.

2 *Disadvantages:* only a sample and doesn't show the whole area. Still takes a considerable amount of time.

3 Line transects will vary, however, should look similar in construction to that of the example.

4 An ecologist would perform several transects of the area to gather as much information as possible to limit any error in the decisions made.

5 Line transects can be useful to get an overview of the layout of the area, so that the construction can be placed in areas that have the least environmental impact.

6 Environmental factors such as rainfall, topography, availability of water etc.

8e Biodiversity

1 **a** *biodiversity:* the difference in range and type of organisms living in an ecosystem. **b** *environment:* all parts of the biota.

c *micro-organism:* small organism, usually observed by magnification. **d** *species:* a group of organisms with common characteristics, which can breed to produce fertile offspring.

e *pollution:* unclean material which destroys the environment.

f *feral:* the result of domestic animals or plants moving into the wild.

2 **a** 25% **b** 10 **c** The environment may change and can no longer sustain a species; introduced predators or man may hunt a species to extinction. **d** The effects would be felt down the natural food chain and would result in dramatic changes to the environment.

3 Increasing populations and urban development are having a greater impact on the environment and causing more species to become extinct.

4 **a** 80% **b** The country was separated early in the evolution process.

5 **a** Answers will vary, should include rabbits, foxes, prickly pear, Queensland cane toad.

b Answers will vary.

8f Losing Our Biodiversity

1 & 2 Answers will vary.

3 **c** A wide range of species ensures a natural balance within the environment. But apart from the moral problems of destroying individual organisms, let alone entire species, just looking at the situation from a scientific and financial viewpoint, we do not yet know what useful chemicals, fuels or other resources could possibly be obtained in the future from all these animals and plants which we ignore, tread underfoot and generally destroy in the name of progress and consumerism.

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